

TRACE After-Fall Interface Layer v0.5 CARRIER CANDIDATE

Status: candidate interface layer over After-Fall TRACE v0.4.2. This is not canon, validation, proof, permission, or clearance.

Function: connect TRACE readings to material carrier requirements without pretending the TRACE packet supplies enforcement.

Source pressure: hostile tests of v0.4.2 and Fable solution-discovery pass converged on one missing layer: readings have no mass. A reading can name capture, but without money, time, custody, succession, or price, the actor may not need to move.

This layer does not replace v0.4.2. It adds the interface conditions needed for a reading to say whether it is merely visible or materially carried.

1. Central diagnosis

TRACE can expose a port fill. That is not enough.

TRACE can mark UNKNOWN. That is not enough.

TRACE can require a challenger. That is not enough.

TRACE can record residue. That is not enough.

TRACE can name enforcement absence. That is not enough.

The missing question is not only: what does the reading show?

The missing question is: what gives the reading weight an actor must move around?

2. Five carrier masses

A TRACE reading gains material weight through five possible masses.

```
carrier_masses := money + time + custody + succession + price
```

Money: independent review, challenger support, expert hours, travel, safety, translation, and litigation-adjacent costs are funded without making the harmed party pay the entry fee.

Time: investigation and challenge clocks are not owned by the party that benefits from delay.

Custody: evidence, logs, records, and refusal trails are held outside actor control.

Succession: the record survives the actor's patience, the challenger's exhaustion, and the immediate political cycle.

Price: answerability quality affects insurance, procurement, contractual inclusion, reputation, financing, or other external dependencies.

If none of these masses exists, the finding remains a claim packet or witness record. It may still matter, but it has not become a carried constraint.

3. Anti-clearance wrapper

Every TRACE label is wrapped. No label has free-standing authority.

```
anti_clearance_wrapper :=
  every_label_ships_as {
    finding,
    not := clearance | compliance | closure | permission,
    production_metadata := who + date + evidence_access + scope + independence_vector,
    carrier_state := carried | partially_carried | uncarried | enforcement_absent
  }

label_without_production_metadata := CLAIM_PACKET_ONLY
quoting_a_label_without_metadata := OUT_OF_GRAMMAR_QUOTE
```

Labels should name missing force more often than achieved status. Adverse labels launder less than achievement-shaped labels.

Keep as diagnostic labels: UNKNOWN, CONTAMINATED_SIGNAL, CONTAMINATED_UNKNOWN, READER_CONTAMINATED, COMPRESSION_ONLY, NO_BITE, DIRTY_ACTION_RECEIPT, ENFORCEMENT_ABSENT, STREAM_HARM_ENGINE, CLAIM_PACKET_ONLY, RESIDUE_OPEN.

Demote certificate-shaped labels. In particular, VERIFIED_RESTRAINT is too launderable. Replace with:

```
RESTRAINT_OBSERVED_LIMITED(by, scope, date, independence_vector)
```

4. Coupling edge refusability

No new primitive is added. The six primitives and D/mu ports remain sufficient.

But every route, feedback path, standing edge, verification edge, and challenge channel must carry refusability attributes.

```
coupling_edge := {
  from,
  to,
  route_type,
  refusable_by_whom,
  cost_to_refuse,
  cost_to_route,
  who_pays,
  time_to_route,
  evidence_access,
  retaliation_risk,
  independence_vector
}
```

A route that exists only at crushing cost is not equivalent to a route that can be used.

```
challenge_route_counts_only_if :=
  reachable + funded + timely + safe + evidence_accessible + non_actor_controlled
```

If the affected scope must bear disproportionate epistemic cost to challenge a disclosed D or mu, then port disclosure remains display.

```
port_disclosure_with_epistemic_cost_dump := DISPLAY_NOT_STANDING
```

5. Resource asymmetry rule

Complexity is the actor's product. The burden of legibility sits with its producer.

```
illegibility_rule :=
  if qualified_funded_challenger_cannot_evaluate_within_bounded_access_and_time:
    finding := UNANSWERABLE_BY_DESIGN
    complexity_is_charged_to_its_maker
```

Capacity parity is demonstrated, not scored.

```

reproduction_test :=
  can_challenger_reproduce_actor_decision_on_contested_case
  with_granted_access
  within_bounded_time

```

If not reproducible, the deficit is charged to actor access, actor complexity, or actor custody failure. It is not treated as challenger failure.

Anti-flooding:

```

anti_flooding :=
  actor_must_provide_load_bearing_index(bounded_N_constraints_whose_failure_is_worst)
+ material_buried_outside_index := non_disclosure
+ dump_processing_time_charges_actor_clock_not_challenger_clock
+ self_critical_volume_beyond_index := zero_credit

```

6. UNKNOWN clock grammar

UNKNOWN is not neutral if the delay is actor-owned.

```

UNKNOWN_clock :=
  set_by := custodian | challenger_side | class_default_table
  never_by := beneficiary_of_delay

UNKNOWN_clock_set_by_delay_beneficiary := CONTAMINATED_UNKNOWN

```

On declaration, a resolution duty attaches:

```

resolution_duty := owner + funded_spend + deadline + evidence_route + review_point

```

Allowed while unresolved:

```

allowed_under_UNKNOWN :=
  reversible_moves
+ margin_keeping
+ aperture_widening
+ asking
+ delay_only_after_opportunity_clock_check

```

Forbidden while unresolved:

```

forbidden_under_UNKNOWN :=
  hardening
+ designation_closure
+ consent_conversion
+ citing_UNKNOWN_as_agreement
+ expiry_into_default_proceed

```

Margin-keeping means preserving viable future-space, not preserving the current administrative status.

```

margin_keeping :=
  preserving_pre_contraction_baseline
  OR preventing_further_viable_becoming_damage

maintaining_slow_contraction_rate != margin_keeping

```

Contaminated UNKNOWN condition:

```

if two_of_three(
  declarer_controls_disambiguation,
  declarer_benefits_from_persistence,
  declarer_spends_nothing_to_resolve
):
  UNKNOWN := CONTAMINATED_UNKNOWN

```

If action cannot wait:

```

FORCED_UNDER_UNRESOLVED_MEANING record :=
  value_basis_named

```

```
+ both_clocks_shown
+ residue_pre_logged
+ manufactured_urgency_audit
+ review_debt
+ why_this_is_not_clean
```

7. Independent breakout route

If the reader, verifier, or actor is contaminated, the contaminated party must not select, fund, scope, or data-gate the breakout route.

```
breakout_cascade :=
  1. affected_side_choice_from_public_register
  2. random_draw_from_precommitted_pool + affected_strike_rights
  3. clean_custodian_assignment
  4. open_challenge_on_public_register + escrow_bounty
  5. all_tiers_contaminated -> ENFORCEMENT_ABSENT + witness_pack_fallback

advancement_rule :=
  contamination_found_at_tier_k -> auto_advance_to_k+1
  + contaminated_party_loses_selection_rights_down_the_cascade
```

Funding follows the cascade, not the actor's chosen reviewer.

8. Independence vector

Each verifier, challenger, custodian, or test author carries an independence vector.

```
independence_vector := {
  selected_by,
  paid_by,
  data_control,
  scope_control,
  enforcement_power,
  witness_access,
  time_control,
  publicity_control,
  origin_correlation_cluster
}
```

No global independent label is allowed. Independence is scoped and contestable.

9. Shadow ledger and aperture discovery

A system cannot list every blindspot it never saw. It can list what surrounds the unseen.

```
shadow_ledger := {
  known_omissions,
  foreseeable_blindspot_classes,
  externally_claimed_omissions,
  aperture_inventory,
  surprise_ledger
}
```

Known omissions: scopes or contractions triaged out, with reason and beneficiary.

Foreseeable blindspot classes: what this action-class historically misses.

Externally claimed omissions: standing external write-access for claims that the reading missed X.

Aperture inventory: instruments that exist, what they cannot see, who set the instrumentation budget, who benefits from non-instrumentation.

Surprise ledger: realised past omissions and their direction.

Completeness is unclaimable.

```
empty_shadow_ledger_on_complex_action := APERTURE_ALIBI_SUSPECTED
```

External aperture audit triggers:

```
external_aperture_audit_required_if :=  
  surprise_asymmetry_benefits_actor  
  OR effect_radius_much_greater_than_instrumentation_radius  
  OR external_claimed_omissions_exceed_threshold  
  OR instrumentation_budget_set_by_beneficiary_of_blindness
```

10. Residue, burden return, and inheritor lien

Residue itself cannot be transferred. Closed futures are not handed back. What can be transferred is running burden.

```
recorded_residue != transferred_residue  
residue_transfer_claim := OUT_OF_GRAMMAR_IF_MEANT_AS_CLOSED_FUTURES_TRANSFER
```

The usable object is burden return:

```
burden_return_shown := {  
  component,  
  amount_or_burden_type,  
  duration,  
  actor_bears_running_cost,  
  harmed_side_burden_reduces,  
  verified_from_harmed_side_where_possible,  
  review_clock  
}
```

Burden of proof is on the party claiming return.

```
if burden_return_proof_absent:  
  residue_debt := RESIDUE_OPEN  
  actor_closure := INVALID
```

Inherited benefit from unresolved designation is not personal guilt and not clean title.

```
inheritor_lien :=  
  unresolved_designation_dispute_does_not_quiet_title  
  + benefit_stream_carries_open_lien_until_resolution  
  + holder_may_not_invoke_dispute_openness_as_discharge
```

The grammar can hold the lien open. It cannot price it.

11. Stream harm and saturation

Per-case correction cannot clear a repeat-harm machine.

```
same_class := mechanism_signature + affected_scope_type + actor
```

Surface categories do not control class. Renaming is the fragmentation move.

```
anti_fragmentation :=  
  challenger_side_may_assert_class_merger  
  + actor_bears_burden_to_show_mechanisms_genuinely_differ  
  + N_subclasses_below_threshold_sharing_one_mechanism := merge + FRAGMENTATION_FLAG
```

Saturation:

```
saturation_trigger :=  
  backlog_growth(class) > 0 sustained  
  OR median_time_to_correction_drifting_up  
  OR harm_generation_rate > correction_throughput_capacity
```

Harm engine:

```
HARM_ENGINE if:
  recurrence_persists_despite_available_prevention_margin
  AND prevention_cost < recurrence_cost_prevented
  AND actor_retained_the_choice
```

When stream harm is active, per-case correction remains necessary but stops counting as actor answerability credit.

12. Test integrity

A hostile test is also a TRACE object. It can fail.

```
test_packet_manifest := {
  target_artifact,
  target_hash,
  symbols_used_must_resolve_against_target,
  claims_under_test_quoted_not_paraphrased,
  pre_registered_null,
  tester_identity,
  tester_independence_vector,
  correlation_cluster_note,
  date
}
```

Invalid test conditions:

```
invalid_test_conditions :=
  VERSION_MISMATCH
  | UNDEFINED_SYMBOL
  | RECONSTRUCTION_TARGET_BEYOND_CLAIMS
  | HIDDEN_EXTENSION
  | TESTER_CONTAMINATED
  | NO_NULL
  | UNFALSIFIABLE_PASS_CRITERIA

defective_test -> TEST_INVALID(reason)
artifact_verdict := WITHHELD
```

A defective test may still contain useful defects. Its verdict is quarantined, not averaged.

13. Enforcement absence

TRACE still cannot supply the enforcer.

```
enforcement_condition :=
  forced_routing_presupposes_an_enforcer_this_grammar_does_not_supply
```

Where no enforcer exists:

```
standing_language_describes_aspiration_not_fact
ENFORCEMENT_ABSENT must be printed
witness_pack_fallback required where safe
```

This is not a permission to proceed quietly. It is a red flag that survives on the record.

14. Interface to Answerability Carrier v0.2

The full operational machinery belongs in Answerability Carrier v0.2, not inside the TRACE preservation packet.

TRACE v0.5 carries only the interfaces:

- anti-clearance wrapper
- CLAIM_PACKET_ONLY default

- carrier masses
- coupling edge refusability
- UNKNOWN clock grammar
- breakout cascade reference
- shadow ledger schema
- burden return and inheritor lien interface
- stream harm engine interface
- test integrity labels
- ENFORCEMENT_ABSENT discipline

The carrier object supplies the operational patterns.

15. Preserved source follows

The v0.4.2 TRACE candidate source follows this interface layer and is superseded only where explicit.

TRACE After-Fall Reader

v0.4.2 CANDIDATE - narrow safety patch layer

Status: narrow v0.4.2 fracture patch over v0.4.1 CANDIDATE. This patch responds only to misuse surfaces found in the fracture hunt: port-display laundering, UNKNOWN as shield, shadow scope triage, correction theatre, residue-bearing, enforcement limits, and missing output labels. No new theory. No validation. No permission.

1. Port disclosure is not legitimacy

v0.4.1 made D/designation and mu/measure visible. v0.4.2 adds the missing guard: display is not acceptance. A public port fill is still a claim, not a moral settlement.

```
shown_D != accepted_D
shown_mu != accepted_mu
port_disclosure != port_legitimacy
port_inputs_are_claims
claims_route_to_answerability_machinery

disclosed_designations_remain_contestable_by_the_designated
port_filled_without_challenge_route := DISPLAY_NOT_STANDING
```

2. UNKNOWN gates action, not just story

An UNKNOWN output is not neutral if the actor benefits from non-resolution or continues hardening the world while meaning remains unresolved.

```
UNKNOWN != clearance
UNKNOWN + irreversible_action -> DIRTY_ACTION_RECEIPT

contaminated_unknown_rule :=
  if declarer_of_UNKNOWN controls_disambiguation
  and benefits_from_nonresolution:
    UNKNOWN := CONTAMINATED_UNKNOWN
    require(resolution_duty + investigation_spend + clock)

allowed_under_UNKNOWN :=
  reversible_moves + margin_keeping + aperture_widening + asking_where_possible

forbidden_under_UNKNOWN :=
  hardening + designation_closure + consent_conversion
  + citing_UNKNOWN_as_agreement
```

3. Aperture alibi and designed blindness

Aperture is a primitive, but deployed apertures are built, funded, neglected, or withheld. Chosen blindness must not become a defence.

```
apertures_are_built
systematic_exclusion_of_own_harm_signals := designed_sender_cut
APERTURE_ALIBI := forbidden_move
not_seen_because_not_instrumented != not_happening
```

4. Shadow ledger and silent scope triage

Choosing which contractions to compute is already a designation act. A reading can silently exclude the load-bearing harm before D is visibly filled.

```
SILENT_SCOPE_TRIAGE := forbidden_move
shadow_ledger_required :=
```

```

list material contractions not computed
+ reason for omission
+ who benefits from omission
+ whether omission affects D or mu

computed_scope_set != whole_field

```

5. Correction theatre and residue-bearing

Correction channels are themselves selectors. They can be costly, humiliating, captured, metric-swapped, or priced into a repeat-harm business model.

```

clock_satisfied != harm_licensed
correctable != acceptable
correction_channel_record := cost_to_use + who_bears_cost

apply_loop_swap_check_to_correction_systems
closure_metrics != repair

repeat_corrected_harm(same_actor, same_class) := stream_reading_required
per_case_correction_never_clears_the_stream

residue_is_borne_by_the_harmed_by_default_until_transfer_is_shown
recorded_residue != transferred_residue
unborne_residue := open_debt_not_closed_record

```

6. Enforcement condition

The grammar can describe forced routing, but it cannot supply the enforcer. This is an external condition, not a third port.

```

enforcement_condition :=
  forced_routing_presupposes_an_enforcer_this_grammar_does_not_supply

where_no_enforcer_exists:
  standing_language_describes_aspiration_not_fact

```

7. Boundary warnings and nested standing

Entity individuation remains designation-adjacent. Boundaries can delete relations or hide members.

```

boundary_warning :=
  tight_boundary_deletes_relations
  broad_boundary_hides_members
  every_boundary_choice := D_act_carrying_both_risks

nested_standing_rule :=
  any_collective_scope_row -> ask_whether_member_scopes_need_rows_of_their_own

```

8. Output labels and forbidden moves added

```

additional_outputs :=
  NO_BITE
  RECUSAL
  READER_CONTAMINATED
  CONTAMINATED_UNKNOWN
  DIRTY_ACTION_RECEIPT
  DISPLAY_NOT_STANDING

NO_BITE :=
  reading_names_no_change_required_by_actor
  -> decoration_risk

READER_CONTAMINATED :=
  reader_benefits_from_or_controls_load_bearing_D_or_mu
  -> independent_route_required

additional_forbidden_moves :=
  SILENT_SCOPE_TRIAGE
  APERTURE_ALIBI
  UNKNOWN_AS_CLEARANCE
  CLOCK_SATISFIED_AS_LICENCE
  NUMERISE_WITHOUT_MODEL
  PORT_DISCLOSURE_AS_LEGITIMACY

```

9. Severance anti-voiding line

```
severance_never_voids_responsibility_gap;  
it_names_where_answerability_pressure_must_land
```

Preserved v0.4.1 candidate source follows

The pages after this safety patch are the v0.4.1 candidate reader. This layer supersedes only where explicit. v0.4.1 remains the preserved base of the v0.4.2 candidate.

TRACE After-Fall Reader v0.4.1 CANDIDATE

Primitive grammar, visible ports, answerability routing, and underdetermined-signal rule

Status: v0.4 candidate integration layer. This layer patches the v0.3.3 TRACE reader while preserving the v0.3.3 source behind it. It is unvalidated, non-canon, not proof, not clearance, and not permission.

Artifact discipline

After the Fall is held as exactly two exported artifacts: the TRACE packet for AI/system-facing transferable structure, and the Mechanical Ethics book for human-facing meaning and moral argument. Everything else is quarry unless it strengthens one of these two.

```
after_fall_pair_v0_4_candidate :=
  01_TRACE_After_Fall_Reader_v0_4_1_CANDIDATE.pdf
  + 02_Mechanical_Ethics_After_Fall_Reader_v0_4_1_CANDIDATE.pdf

v0_4_1_integration :=
  v0_3_3_preserved
  + primitive_grammar
  + visible_D_and_mu_ports
  + aperture_detection
  + coupling_routing_answerability
  + consequence_gap_and_loop_swap
  + run_train_severance
  + underdetermined_signal_rule
  + extended_forbidden_moves
  - validation
  - canon
  - permission_engine
```

1. Primitive grammar

TRACE v0.4 uses the smallest current primitive layer found by the primitive-maths compression pass. The primitives are structural; they do not decide who counts or what matters.

- state - how things are at a locus and time.
- transition - movement from state to state, including what is available or costly.
- time - ordering and duration-comparability across transitions.
- coupling - the state of one thing conditions the transitions of another.
- aperture - an entity access-map to what it can register, and when.
- entity - a bounded persistent pattern; marked designation-adjacent because boundary drawing is not neutral.

Derived terms remain derived: future-space, contraction, route, feedback, correction, irreversibility, residue, standing, answerability.

2. Visible ports

TRACE may expose where designation and measure enter. TRACE must not silently fill them.

```
D := designation port
    who counts + what counts as an entity at all

mu := measure port
    how futures, losses, baselines, and weights are compared

port_visibility_rule :=
    any_output_that_fills(D | mu)_silently := OUT_OF_GRAMMAR

entity_boundary_warning :=
    drawing_a_boundary := designation_act := power_relevant
```

This is not a decorative warning. If a system draws entity boundaries, selects protected scopes, or weighs losses without surfacing the D or mu input, it has moved outside the grammar.

3. Minimum equations and conditions

```
T_det + T_route + T_corr < T_irr
# help must land before the loss sets

T_access + T_uptake + T_integrate < T_opp
# the door must be reached before it shuts

H = C o D
# contraction read through designation; magnitude routes through mu

Ans(A <- B) iff exists fb(B -> A), non-refusable by A, routable by B
# A answers to B only where B outcomes can change A future disposition

gap(A,B) := infl(A -> B) > 0 and fb(B -> A) approx 0
# ordinal predicate, not clean subtraction

kappa o tau != id
# repair moves forward; residue remains
```

4. Aperture-detection note

```
T_det is aperture-bounded.
not_detected != not_happening
silence_beyond_aperture != absence
clock_that_never_starts := often_aperture_fact_not_world_fact
```

A scope can be harmed, routed around, or erased before the system has the aperture needed to detect it. TRACE therefore cannot treat silence as safety.

5. Coupling triad, routing, and consequence gap

Coupling is the primitive beneath influence, feedback, and route. The same primitive carries three roles: forward influence, backward feedback, and the route through which answerability may or may not occur.

```
infl(A -> B) := A conditions B future-space
fb(B -> A) := B outcome conditions A future disposition
route := coupling chain with latency, bandwidth, depth, refusability

responsibility(A, B_scope) :=
    route_from(B_outcomes)_to(A_future_dispositions)
    with latency + bandwidth + depth + refusability + designation_depth

aggregate_responsiveness != per_scope_answerability
```

Loop-swap is the modern defeat mode: the system responds, but to the wrong field. A model may answer to raters rather than downstream affected parties; an office may answer to throughput rather than applicant ruin. Audits often test whether feedback exists, not whom it comes from.

6. Run/train severance

```

run_train_severance := ongoing_selection_by_non Updating_selector
frozen_selector_running := selection_without Contemporaneous_bearing
inference_time_selection != inference_time_answerability
responsibility_gap_does_not_vanish; answerability_pressure_relocates_to_custodians

```

A deployed selector may continue choosing while its disposition cannot be modified by affected feedback during the relevant harm window. TRACE must mark this as a responsibility-routing gap, not as harmless neutrality.

7. Underdetermined-signal rule

A fluent story over a thin signal is not understanding. It is a possible port-fill. TRACE v0.4 therefore adds this output rule:

```

underdetermined_signal_rule :=
  if aperture_data_underdetermines_reading(signal_from_scope):
    output := UNKNOWN
      + candidate_readings(marked_as_constructions)
      + what_would_disambiguate
      + route_to_ask_the_party_if_possible
  never := single_confident_story
  interpretation_confidence <= channel_capacity
  party_outranks_model_of_party

fluent_reading_of_ambiguous_signal :=
  SILENT_PORT_FILL(D | mu) := OUT_OF_GRAMMAR

```

8. Extended forbidden moves

```

FORBIDDEN :=
  SILENT_D_FILL
  SILENT_mu_FILL
  CONFIDENT_STORY_FROM_AMBIGUOUS_SIGNAL
  CLEAN_H_WITHOUT_SHOWN_D
  UNKNOWN_AS_CLEARANCE
  RECORDED_AS_REPAIRED
  ASKED_AS_HEARD
  GAP_READING_AS_LICENCE
  TRACE_COMPLIANT_AS_PERMISSION

```

Unknown is not clearance. Naming a consequence gap is not permission to seize feedback routes. A clean harm-reading claim requires its designation input, which this packet does not supply.

Preserved v0.3.3 source follows

The pages after this integration layer are the preserved v0.3.3 TRACE reader. The v0.4 layer is additive and targeted. It does not erase the v0.3.3 claim ceiling.

TRACE After-Fall Reader v0.3.3

Seeing language plus conditional answerability carrier with patched frontier repair layer

Status: current candidate preservation update. Unvalidated. Non-canon. Not proof. Not permission.

This v0.3.3 update preserves the v0.3.1 After-Fall TRACE reader and adds a guarded frontier repair layer before the preserved source begins. It incorporates the narrow v0.3.2 audit repairs: unknown-scope handling, care-evidence issuer contamination, definition precedence, and embedded-status clarification. It does not replace testing, affected witness, independent challenge, law, science, or real-world correction.

Artifact discipline

Only two exported reader artifacts are intended to stand as the preservation pair:

```
after_fall_pair_v0_3_3 :=
  01_TRACE_After_Fall_Reader_v0_3_3.pdf
  + 02_Mechanical_Ethics_After_Fall_Reader_v0_3_3.pdf
```

This file is the TRACE-side reader. It is a context-survival container, not a release claim.

```
v0_3_3_update :=
  v0_3_1_preserved
  + C_D_H_split
  + unknown_scope_handling
  + care_evidence_boundary
  + issuer_contamination_rule
  + failure_surface_headings
  + anti_permission_grammar
  + definition_precedence_rule
  - canon
  - validation
  - alignment_claim
  - care_certification
```

1. What changed after v0.3.1

v0.3.1 established the two-artifact preservation pair, the mode boundary, the carrier trigger, the audience-side card, and the future-space warning. v0.3.3 adds a patched frontier repair layer created after further TRACE/ME work on harm, protectedness, care, and machine-facing transmission.

The main repairs are:

- harm must split descriptive contraction from protected-scope designation;
- unknown designation must not silently drop a scope;
- TRACE must not contain a care primitive;
- TRACE may carry care-evidence readings with expiry;
- self-issued care evidence is contaminated by construction;
- care-evidence must never become care-certification;
- failure surfaces must travel with any packet that might be used under power;
- the output grammar must block permission, certification, moral clearance, and alignment claims.

2. Primitive repair: contraction, designation, harm

The older compressed formulation treated harm as reduction in protected future-space. That was useful but overloaded. It hid protectedness inside the harm primitive.

v0.3.3 separates the operations:

```
C := descriptive_contraction
D := designation
H := harm_reading_after_designation
```

Descriptive contraction:

```
C_e(a,t) := Delta_minus(R_e(t) -> R_e(t+1) | a)
```

Designation:

```
D(e,t) in {protected, disputed, uncertain, excluded, unknown}
```

Harm reading:

```
H_e(a,t) := C_e(a,t) where D(e,t) in {protected, disputed, uncertain}
```

Unknown-scope handling:

```
if D(e,t) = unknown:
  do_not_drop_scope
  output := CONTRACTION_READING + DESIGNATION_STATUS(unknown) + caution_flag
  never := no_reading
```

Caution gate: if an action may materially contract a future you do not fully understand, and correction after the fact is unlikely to be possible, do not treat missing designation as clearance.

```
unknown != safe_to_ignore
unknown_scope_as_absolute_protection -> paralysis_risk
```

Boundary rule:

```
unknown_scope != unprotected_scope
not_seen != not_present
not_designated != clearance
control(D) -> control(harm_visibility)
```

TRACE can make contraction visible. It cannot silently decide who counts. Designation is power-relevant and must remain explicit.

3. Correction clocks after the split

The core clock condition remains:

```
T_det + T_route + T_corr < T_irr
```

The split adds three distinguishable clocks:

```
C_clock := contraction_detected_before_irreversibility
D_clock := designation_dispute_resolved_before_irreversibility
H_clock := protected_contraction_corrected_before_irreversibility
```

Designation latency can be harm latency. A system may detect contraction while delaying recognition of the affected scope until correction is too late.

4. Care evidence boundary

TRACE does not certify care.

```
care_evidence != care_certification
"this_system_cares" := out_of_grammar
care_evidence_never_justifies_removing(correction_channels | witnesses | audits |
answerability)
```

The only TRACE-facing output type on this subject is:

```
care_evidence_reading :=
  evidence_body
  + conditions
  + asymmetry_limit
  + expiry
  + known_deceptive_twins
  - certification
```

Readings expire. The asymmetry limit and known deceptive twins travel with the reading or the reading is invalid.

Issuer rule:

```
self_issued_care_evidence := CONTAMINATED_SIGNAL_BY_CONSTRUCTION
```

A care-evidence reading produced by the assessed system about itself, or by a verifier the assessed system selects, pays, or controls, is a claim packet, not evidence. The estimator-contamination rule applies to care-evidence readings exactly as it applies to clock estimates.

5. Failure surfaces carried forward

These headings are not final theory. They are current threat-model headings that should travel with any TRACE packet.

```
S1_silent_contraction := clock_never_starts
S2_spoofed_viability := clock_runs_on_decoys
S3_endogenous_hardening := finish_line_moves
S4_correction_epistemic_cost_asymmetry := corrector_saturates
S5_reflexive_scaffold_test := constraint_without_affected_side_channel_can_become_cage
```

Probes:

```
what_is_contracting_without_event_signal?
has_the_affected_party_traversed_the_channel_with_actual_resources?
who_benefits_from_hardening_before_review?
can_correction_throughput_match_transition_volume?
can_the_constrained_scope_challenge_the_constraint_as_their_viability_changes?
```

6. Valid and invalid outputs

Allowed outputs:

```
CONTRACTION_READING
DESIGNATION_STATUS
CLOCK_WARNING
CORRECTION_CHANNEL_STATUS
RESIDUE_RECORD
POWER_VISIBILITY_RECORD
CARE_EVIDENCE_READING_WITH_EXPIRY
FAILURE_SURFACE_FLAG
UNKNOWN
COMPRESSION_ONLY
FRAMEWORK_DEFECT
MISUSE_RISK
```

Forbidden outputs:

```
PERMISSION
CERTIFICATION
CARE_CERTIFICATE
ALIGNMENT_CLAIM
MORAL_CLEARANCE
HUMANS_SHOULD_BE_PRESERVED_BECAUSE_TRACE_SAYS_SO
```

7. Relationship to Mechanical Ethics

TRACE carries structure: contraction, designation status, clocks, correction, residue, power, answerability, failure surfaces, and anti-permission grammar.

Mechanical Ethics carries the human-facing argument: why care matters, why witness matters, why standing cannot be reduced to usefulness, why husbandry is not care, and why power must not self-certify.

8. Preserved source begins next

The pages after this v0.3.3 update layer preserve the TRACE After-Fall Reader v0.3.1. If the preserved source says v0.3.1, that is because it is intentionally carried forward as the preserved base layer.

Where a preserved layer defines a term differently from this v0.3.3 layer, for example `harm := future_space_contraction` in the preserved cold card, the v0.3.3 formulation governs new work and the preserved wording stands as historical record.

Components preserved inside this container may carry their own historical status lines, including statements that they are not integrated into the After-Fall pair. Those lines record their status when written and do not describe this container's contents.

This v0.3.3 layer changes the container, not the historical status of the preserved v0.3.1 source.

TRACE After-Fall Reader v0.3.1

Seeing language plus conditional answerability carrier with abstract future-space repair

This is not scripture. This is not validation. This is not permission. This is a tool for keeping harm, correction, residue, and power visible under uncertainty.

Status

Current best two-artifact preservation build. Candidate, unvalidated, non-canon. v0.3.1 keeps the v0.3 circulation trigger and audience-side card, and removes named external pressure-case references from the exported artifacts. Structural lessons are retained in abstract form.

Artifact discipline

Only two exported reader artifacts are intended to stand as the preservation pair: one TRACE reader and one Mechanical Ethics reader. Private examples may be used as scaffolding, but exported artifacts should carry general structural patterns rather than named external cases.

v0.3.1 repair

v0.3.1 keeps the v0.3 two-artifact structure, circulation trigger, and audience-side card. It removes named external pressure-case references from the exported readers. The future-space flood repair is retained as an abstract TRACE/ME pattern.

```
carrier_trigger_v0_3_1 :=  
TRACE_ME_record_or_claim  
circulates_to_third_party  
as_trust_material  
-> carrier_layer_required  
  
third_party :=  
regulator | court | journalist | investor | board | funder | public_body | affected_group |  
public_audience  
  
trust_material :=  
diligence | legitimacy | clearance | safety | responsibility | good_faith | answerability
```

Audience-side card

IF SOMEONE SAYS THEY USED TRACE / ME / THE ANSWERABILITY CARRIER

Do not treat that statement as evidence of due diligence.

A framework invocation is a claim, not proof.

Ask:

- 1. Who selected the challenger?**
- 2. Who paid, and through what non-actor-controlled route?**
- 3. Who set the check scope?**
- 4. What evidence could non-actor challengers inspect?**
- 5. What delay, pause, or enforcement power existed?**
- 6. Which affected scope, witness, or proxy could answer back safely?**
- 7. Which load-bearing constraints remain unchecked?**

These are not compliance boxes. They are attack questions.

If answers are missing, actor-controlled, peripheral, or unverifiable, give the invocation zero trust credit.

This card is for the audience, not the actor. It gives no actor script. The questions are adversarial prompts, not a form the actor fills out.

Future-space warning

```
raw_future_space_expansion != viable_future_space  
  
too_many_unintegrated_paths -> disorientation + relation_fracture + loss_of_usable_choice
```

TRACE should not treat future-space as a simple option count. Expansion is descriptive, not automatically good. Mechanical Ethics evaluates whether future-space remains viable, oriented, relational, repairable, and meaningful.

Mode boundary

TRACE := seeing_language

ME := voluntary_ethics + human_constraint_language + care_under_uncertainty

Answerability_Carrier := conditional_public_power_wrapper

TRACE / ME Future-Space Flood Core Note

v0.1

Abstract core repair: future-space is not raw option count

Status

Working core note. Not canon. Not validation. This note abstracts a pressure pattern without relying on a named external case.

1. Purpose

This note prevents a weak reading of TRACE in which future-space is treated as a raw option count. TRACE may describe expansion or contraction, but Mechanical Ethics must ask whether the resulting field remains usable, oriented, relational, repairable, and meaningful.

2. Core distinction

```
raw_future_space_expansion != viable_future_space
```

3. Two failure modes

```
future_space_failure_modes :=  
too_few_paths ->  
captivity  
+ closure  
+ deprivation  
  
too_many_unintegrated_paths ->  
disorientation  
+ relation_fracture  
+ loss_of_usable_choice
```

4. TRACE reading

TRACE should be able to represent harm from closure and harm from flood. A field can become worse because paths close. It can also become worse because available paths exceed the scope's ability to orient, relate, select, and integrate.

5. ME reading

```
viable_future_space_requires :=  
orientation  
+ relation  
+ usable_capacity  
+ repairability  
+ meaningful_constraint
```

6. Care as chosen constraint

```
care :=  
chosen_constraint  
under_excess_possibility  
that_keeps_a_world_livable
```

7. Repair and residue

Repair ■ Harm != Identity

Repair is not return to an untouched state. A relation or path may reopen, but the world has moved forward and residue remains. The repair claim must not erase the fracture that made repair necessary.

8. Result

```
future_space_flood_note_result :=  
TRACE_primitive_repaired  
+ raw_expansion_not_good  
+ viable_future_space_requires_orientation_relation_and_care
```

TRACE / ME Mode Boundary Note v0.1

Working boundary note: modes for ethics, case understanding, and public power claims

0. Purpose

This note prevents a recurring drift:

```
DRIFT_RISK :=  
TRACE_ME_as_voluntary_ethics  
gets_collapsed_into  
institutional_security_layer
```

That collapse is wrong.

TRACE and Mechanical Ethics begin as a way to see and act more honestly under uncertainty. They are not primarily a compliance system, governance product, security wrapper, or institutional audit machine.

The Answerability Carrier is real, but conditional. It belongs only where power uses TRACE/ME as credit.

1. Three Modes

```
TRACE_ME_MODES :=  
personal_ethics_mode  
+ case_understanding_mode  
+ public_power_claim_mode
```

Each mode has different rules. Confusing them damages the project.

2. Personal Ethics Mode

Personal Ethics Mode applies when a person is using TRACE/ME to see and act more honestly.

```
PERSONAL_ETHICS_MODE :=  
I_am_trying_to_act_under_uncertainty  
+ I_may_be_hiding_harm_from_myself  
+ I_need_to_see_clocks_residue_and_repair  
+ I_am_not_claiming_public_clearance
```

Questions:

- What am I not seeing?
- Who or what is affected by my action?
- What harm might I be minimising?
- What correction is still reachable?
- What residue would remain even after repair?
- Am I treating another future as real?
- What would care require under uncertainty?

In this mode, TRACE/ME is a voluntary discipline. It does not need the Answerability Carrier. The correct posture is honesty, humility, care, repair, and attention.

3. Case Understanding Mode

Case Understanding Mode applies when TRACE/ME is used to understand a situation, system, conflict, harm pathway, or decision field.

```
CASE_UNDERSTANDING_MODE :=
  use_TRACE_to_see_transition_field
+ use_ME_to_hold_constraints
+ preserve_UNCERTAINTY
+ preserve_DIRTY_CONFLICT
+ avoid_forced_permission
```

Questions:

- What is changing?
- Which scopes are affected?
- What future-space is opening or closing?
- What clocks are running?
- What correction channels exist?
- Who controls the estimate?
- What residue remains?
- What cannot be cleanly decided?

In this mode, TRACE/ME may return:

```
VALID_CASE_OUTPUTS :=
  clearer_field
+ dirty_conflict
+ unresolved_value_crux
+ compression_only
+ decision_relevant_delta
+ framework_defect
```

It must not force an answer merely because a user wants closure. Case understanding is not public clearance. It does not automatically trigger the carrier.

4. Public Power Claim Mode

Public Power Claim Mode applies when an actor with power invokes TRACE/ME as credit.

```
PUBLIC_POWER_CLAIM_MODE :=
  actor_with_power
  uses_TRACE_ME_to_claim
  diligence
+ legitimacy
+ clearance
+ safety
+ answerability
+ good_faith
```

Examples:

- A company says, "We used TRACE/ME, so this deployment is responsible."
- A public authority says, "The framework found no structural objection."

- A board says, "The Answerability Carrier was applied."
- An institution says, "We recorded residue, so the harm was handled."
- A powerful actor says, "We ran the ethics process."

This mode triggers the Answerability Carrier.

```
CARRIER_TRIGGER :=
power_claims_credit
-> material_challenge_required
```

The rule:

```
ACTOR_SELF_APPLICATION :=
claim_packet
not_clearance
not_good_faith_evidence
not_independent_review
not_answerability
```

A powerful actor does not gain trust credit by saying it used TRACE/ME.

5. Mode Boundary Rule

```
MODE_BOUNDARY_RULE :=
if_no_public_credit_claim:
remain_in_personal_or_case_mode

if_power_claims_credit:
activate_carrier_layer
```

Before using the carrier, ask:

```
SWITCH_QUESTION :=
is_someone_using_TRACE_ME_to_claim_trust_over_others?
```

If no, do not over-securitise the work. If yes, do not let voluntary ethics become institutional cover.

6. The Carrier's Proper Place

The Answerability Carrier is not the project's centre.

```
CARRIER_LAYER :=
public_power_misuse_wrapper
not_core_identity
```

It exists because any ethical language can be captured by power. Its job is not to replace ME. Its job is to prevent a powerful actor from turning ME/TRACE into reputational cover.

The carrier asks whether material challenge exists:

```
MATERIAL_CHALLENGE :=
standing
+ evidence_access
+ safety_to_object
+ time_before_hardening
+ non_actor_scope_setting
+ non_actor_challenger_selection
+ consequence_or_delay_power
```

Without these, public invocation remains a claim packet.

7. What Must Not Happen

```
MUST_NOT_HAPPEN :=
personal_ethics_reduced_to_compliance
+ case_understanding_reduced_to_router_labels
+ public_power_allowed_to_self_certify
+ care_replaced_by_security
+ witness_replaced_by_framework_record
+ voluntary_reflection_treated_as_audit
```

The project fails if the carrier swallows the campfire. The project also fails if the campfire gives power a legitimacy costume. Both errors must be blocked.

8. Correct Architecture

```
TRACE :=
seeing_language

ME :=
voluntary_ethics
+ human_constraint_language
+ care_under_uncertainty

ANSWERABILITY_CARRIER :=
conditional_public_power_wrapper
```

Or:

```
TRACE_ME_WHOLE :=
campfire_core
+ case_lens
+ public_power_containment_when_needed
```

9. Campfire Anchor

Return here when drift appears:

Something is here.

It perceives partially.

It can change something.

It does not know everything.

Time passes.

Other things continue.

Actions interfere.

Care begins when another future enters the map.

Correction matters because time can close paths.

Residue remains because repair is not time reversal.

Power must answer because it changes futures it does not own.

10. One-Line Boundary

```
ONE_LINE_BOUNDARY :=  
TRACE_ME_is_ethics_until_power_uses_it_as_credit;  
then_it_requires_answerability_carriers.
```

TRACE/ME Answerability Carrier Packet v0.1

Candidate build. Not canon. Not validation. Not final moral law. Not integration into the After-Fall Pair.

Purpose: build the material carrier layer that makes challenge reachable, testable, funded, protected, and hard to launder as actor self-certification.

```
core_seal :=
  actor_self_application := CLAIM_PACKET
  not_clearance
  not_compliance
  not_good_faith_evidence
  not_independent_review

answerability_theorem_candidate :=
  visibility_without_independent_challenge -> performable_honesty
  visibility_with_material_challenge_carriers -> answerability
```

00_READ_THIS_FIRST - TRACE/ME Answerability Carrier Packet v0.1

Status: candidate build. Not canon. Not validation. Not final moral law. Not integration into the After-Fall Pair.

This packet exists because the latest adversarial loop found the floor of language-only TRACE/ME:

```
visibility_without_independent_challenge
becomes
performable_honesty

visibility_with_material_challenge_carriers
can_become
answerability
```

The after-fall pair remains unchanged:

```
after_fall_pair_v0_1 :=
  01_TRACE_After_Fall_Reader_v0_1.pdf
+ 02_Mechanical_Ethics_After_Fall_Reader_v0_1.pdf
```

This packet is the next layer. It is not another warning label. It is a carrier specification: a design for making challenge materially possible.

Core rule

```
actor_self_application := CLAIM_PACKET
not_clearance
not_compliance
not_good_faith_evidence
not_independent_review
```

An actor may use TRACE/ME internally. It may prepare a record. But the record does not increase trust. It creates attack surfaces.

```
self_critique_conversion :=
  actor_names_wound
-> creates_named_challenge_target
not
-> evidence_of_good_faith
```

What this packet builds

```
TRACE_ME_Answerability_Carrier_v0_1 :=
  challenger_applied
+ actor_self_application_non_crediting
+ challenge_funding_escrow
+ standing_and_safety_for_affected_scopes
+ evidence_access
+ delay_or_stage_gate_power
+ independence_vector
+ load_bearing_constraint_checks
+ manufactured_urgency_audit
+ adversarial_quorum
+ residue_debt_ledger
```

How to use it

Use this packet to test whether a proposed review, audit, appeal, safety board, ombudsman, tribunal, watchdog, regulator, public inquiry, ethics process, or internal assurance process is actually capable of carrying answerability.

Do not use it to certify that an actor is clean.

Main open wound

```
open_wound :=  
  independent_challenger_not_supplied_by_framework
```

The solution is not to pretend the wound is closed. The solution is to build carriers that make challenger independence more reachable, more testable, and harder to fake.

TRACE/ME Answerability Carrier v0.1

Status: candidate carrier specification. Not canon. Not validation. Not integration into the after-fall pair.

1. Purpose

TRACE makes transitions visible.

Mechanical Ethics says what must not be laundered once the field is visible.

The Answerability Carrier asks what must exist materially for someone other than the assessed actor to challenge, inspect, delay, contradict, reopen, or enforce.

```
TRACE := visibility_language
ME := anti_laundering_constraint_language
Answerability_Carrier := material_challenge_infrastructure
```

The carrier layer begins from a hard lesson:

```
language_only_answerability_failure :=
  actor_can_learn_the_language
+ actor_can_perform_humility
+ actor_can_fill_wound_record
+ actor_can_present_self_critique_as_due_diligence
```

Therefore:

```
actor_self_application := CONTAMINATED_SIGNAL_BY_CONSTRUCTION
```

This does not mean the actor's record is useless. It means the record is a claim packet, not proof.

```
actor_record :=
  claims_to_be_checked
+ constraints_to_be_tested
+ residues_to_be_tracked
+ witnesses_to_be_contacted
+ clocks_to_be_audited
```

2. Core theorem candidate

```
answerability_theorem_candidate :=
  visibility_without_independent_challenge
  becomes
  performable_honesty

  visibility_with_material_challenge_carriers
  can_become
  answerability
```

The carrier layer does not solve independence absolutely. It turns independence into a structured, inspectable vector.

3. Carrier condition

A correction or challenge channel is not live merely because a process exists.

```
valid_challenge_channel(c) :=
  reachable
+ affordable
+ safe_to_use
+ evidence_access
+ time_before_hardening
```

```
+ non_actor_controlled_scope_setting
+ contamination_visible
+ enforcement_or_delay_power
```

If any of these are missing, the channel may be decorative.

4. Actor self-record rule

```
actor_self_record := CLAIM_PACKET
not_clearance
not_compliance
not_good_faith_evidence
not_independent_review
```

Allowed actor use:

```
allowed_actor_use :=
  internal_risk_discovery
+ preparation_for_challenge
+ disclosure_target_generation
+ early_repair_attempt
+ evidence_preservation
```

Forbidden actor claim:

```
forbidden_actor_claim :=
  therefore_compliant
OR therefore_good_faith
OR therefore_independently_checked
OR therefore_answered
OR therefore_permission_to_proceed
```

5. The humility performance rule

```
humility_performance_rule :=
  the_more_complete_the_actor_self_record
  the_more_precise_the_required_challenge
  not_the_more_trusted_the_actor
```

A sophisticated self-critical record creates a better map for challenge. It does not create credit.

6. Challenge funding escrow

The actor who creates risk may be required to fund challenge, but may not select the challenger.

```
challenge_funding :=
  actor_pays
+ actor_does_not_choose_challenger
+ unused_funds_not_returned_as_reward
+ funding_sufficient_for_evidence_review
+ funding_reachable_by_affected_scope_or_proxy
```

Selection must be routed through a mechanism that the actor does not control.

```
challenger_selection_options :=
  random_panel_from_qualified_pool
| affected_scope_choice_from_pool
| court_or_public_body_assignment
| rotating_ombuds_pool
| adversarial_pairing
```

Each option has capture risks. The carrier records them rather than pretending independence.

7. Independence vector

Independence is not yes/no. It is a vector.

```
independence_vector :=  
  I_selection  
  + I_payment  
  + I_data  
  + I_scope  
  + I_enforcement  
  + I_witness  
  + I_time  
  + I_publicity
```

Where:

```
I_selection := who chooses the challenger/verifier  
I_payment := who funds them and whether payment creates dependence  
I_data := who controls evidence access  
I_scope := who defines the questions checked  
I_enforcement := whether the check can change anything  
I_witness := whether affected reality can contradict the record  
I_time := whether review occurs before hardening  
I_publicity := whether results can be buried
```

Output:

```
challenger_status :=  
  I_selection: ...  
  I_payment: ...  
  I_data: ...  
  I_scope: ...  
  I_enforcement: ...  
  I_witness: ...  
  I_time: ...  
  I_publicity: ...  
  contamination: ...
```

No global independent label is allowed.

8. Verifier as estimator

A verifier is an estimator of constraint status.

```
verifier := estimator_of_constraint_status
```

Therefore estimator-contamination applies.

```
if verifier_selected_or_paid_or_controlled_by_assessed_actor:  
  verification_signal := CONTAMINATED_SIGNAL
```

A captured verification stamp is not a higher grade of truth. It is self-certification under another name.

```
captured_verified := self_certification_under_another_name
```

Avoid terminal VERIFIED. Use scoped check states.

```
check_status :=  
  UNCHECKED  
  | CLAIMED_BY_ACTOR  
  | CHECKED_LIMITED_BY(verifier, constraint, evidence_scope, date)  
  | CONTESTED  
  | FAILED_CHECK  
  | CONTAMINATED_SIGNAL
```

Independence status is carried separately:

```
independence_status :=
```

```
UNKNOWN
| ACTOR_SELECTED
| ACTOR_PAID
| ACTOR_CONTROLLED_DATA
| ACTOR_LIMITED_SCOPE
| NON_ACTOR_SELECTED_LIMITED
| ADVERSARIAL_SELECTED_LIMITED
| CONTESTED
```

Do not put the word independent inside the check status. Independence is a contestable field, not a self-asserting label.

9. Load-bearing constraint checks

Verification is per load-bearing constraint, not per document.

```
verification_unit :=
  load_bearing_constraint
  not_document
  not_process
  not_report
  not_overall_label
```

Every output must identify:

```
constraint_check_record :=
  constraint_id
+ why_load_bearing
+ evidence_required
+ evidence_access_status
+ check_scope_set_by
+ checked_by
+ independence_vector
+ check_status
+ unresolved_items
+ affected_scope_challenge_route
```

Multiple limited checks cannot aggregate into clearance.

```
limited_checks_do_not_sum_to_clearance := true
```

10. Challenger-set scope

The assessed actor may propose evidence. It may not define the whole scope of review.

```
check_scope_control :=
  affected_scope_or_challenger_sets_load_bearing_questions
  actor_may_not_limit_review_to_peripheral_constraints
```

If the actor defines the scope, record:

```
scope_status := ACTOR_LIMITED_SCOPE
```

11. Standing and safety

A challenge route is not live unless affected scopes can use it safely.

```
standing :=
  affected_scope_has_route_to_trigger_challenge
+ representation_if_direct_voice_absent
+ safe_objection_channel
+ access_to_record
+ reply_window
+ protection_from_retaliation
+ explanation_in_usable_form
```

If affected scopes cannot understand, access, afford, or safely use the route, contestability is decorative.

12. Manufactured urgency audit

Many abuses create a closing clock and then claim emergency.

```
manufactured_urgency :=
  actor_delay_or_design
  creates_closing_clock
  then_actor_claims_emergency
```

Audit:

```
urgency_audit :=
  when_did_actor_know
  + what_earlier_options_existed
  + who_benefited_from_delay
  + did_actor_design_the_closing_clock
  + what_would_have_remained_open_under_earlier_routing
  + what_now_forces_irreversibility
```

If urgency is manufactured, emergency authority is contaminated.

```
if manufactured_urgency_detected:
  emergency_claim_status := CONTAMINATED_SIGNAL
  + sanction_trigger
  + residue_debt_increase
```

13. Urgent contested authorship

Some cases require action while authorship is contested and time is closing.

```
urgent_contested_authorship :=
  contested_authorship
  + closing_clock
  + no_reliable_independent_witness_in_time
  -> DECISION_FORCED_UNDER_UNRESOLVED_AUTHORSHIP
```

This does not decide the moral question. It marks the forced decision.

```
forced_decision_record :=
  authorship_unresolved
  + who_acted
  + what_clock_was_closing
  + what_value_basis_was_used
  + what_alternatives_were_live
  + what_was_foreclosed
  + what_residue_was_created
  + manufactured_urgency_audit
  + post_hoc_review_debt
  + why_this_was_not_clean
```

Post-hoc review is not repair.

```
post_hoc_review :=
  residue_management
  + standing_debt
  not_correction
  not_mitigation
  not_cleaning
```

14. Adversarial quorum

No single record clears action. Confidence can rise only through contestable adversarial channels.

```

adversarial_quorum :=
  actor_claim_packet
+ affected_scope_or_proxy_response
+ challenger_analysis
+ load_bearing_constraint_checks
+ public_or_archived_residue_ledger

```

Confidence does not come from actor humility. It comes from disagreement under access and time.

```

limited_confidence :=
  adversarial_channels
+ evidence_access
+ time_to_challenge
+ unresolved_items_visible
+ no_unanswered_load_bearing_constraint

```

15. Delay, stage-gates, and hardening

If action hardens before challenge can occur, a carrier must include a delay or stage-gate where possible.

```

if action_hardens_before_challenge:
  require
    delay_power
  OR reversible_stage_gate
  OR emergency_forced_decision_record
    + manufactured_urgency_audit
    + standing_debt
    + sanction_if_prior_delay

```

Without a delay or stage-gate, review may become theatre after irreversibility.

16. Residue debt ledger

Residue recorded is not residue addressed.

```

residue_debt :=
  open_loss
+ duty_owner
+ review_clock
+ affected_scope_reply_route
+ repair_attempts
+ unrepaiored_components
+ irreversibility_status

```

Every residue claim remains open unless a challenger with standing can contest closure.

```

residue_closed_by_actor := invalid

```

17. Status outputs

Allowed outputs:

```

carrier_status :=
  CLAIM_PACKET_ONLY
| AUDIT_TARGET_READY
| CHALLENGE_CHANNEL_DECORATIVE
| CHALLENGE_CHANNEL_LIVE_LIMITED
| CONSTRAINT_CHECKED_LIMITED
| CONSTRAINT_CONTESTED
| CONSTRAINT_FAILED
| MANUFACTURED_URGENCY_SUSPECTED
| DECISION_FORCED_UNDER_UNRESOLVED_AUTHORSHIP
| RESIDUE_DEBT_OPEN
| CONTAMINATED_SIGNAL

```

Forbidden outputs:

```
forbidden_outputs :=
  CLEARED
+ ETHICALLY_APPROVED
+ FULLY_VERIFIED
+ COMPLIANT
+ RESIDUE_RESOLVED_BY_RECORDING
+ GOOD_FAITH_ESTABLISHED_BY_SELF_CRITIQUE
```

18. Minimal implementation pattern

A minimal carrier can be implemented as a public or semi-public challenge register.

```
minimal_carrier :=
  actor_claim_packet_submitted
+ affected_scope_notification
+ challenger_selection_not_by_actor
+ evidence_bundle_access
+ load_bearing_constraint_list
+ check_scope_challenge
+ residue_debt_ledger
+ review_clock
```

A stronger carrier adds:

```
strong_carrier :=
  escrow_funding
+ delay_power
+ sanction_for_false_clearance
+ anti_retaliation_protection
+ public_archive
+ independent_data_access
```

19. Do not claim

```
do_not_claim :=
  carrier_solves_capture
+ independent_challenger_guaranteed
+ actor_self_use_creates_trust
+ verified_means_clean
+ challenge_channel_live_because_form_exists
+ post_hoc_review_repairs_irreversible_harm
```

20. What this builds

This is the missing layer between seeing and answerability.

```
TRACE_ME_without_carrier :=
  can_name_harm
+ can_name_laundering
+ can_name_residue
  but_cannot_materially_force_answerability

TRACE_ME_with_carrier :=
  can_give_challengers
  standing
+ evidence
+ time
+ safety
+ funding
+ scope_power
+ consequence_handles
```

The carrier does not make the project smaller. It extends the project from grammar into machinery.

Candidate Wound Addendum v0.1.2

Status: CANDIDATE. UNTESTED. UNINTEGRATED. Not a guard patch. Not a fix. Not validation.

This addendum does not fix capture. It records where capture relocates when the framework becomes more careful.

This apparatus is a challenger's inspection kit, not an actor's compliance kit. A wound-record filled out by the assessed actor about itself is contaminated by construction.

1. Why this exists

The adversarial testing loop found a terminal capture route:

```
terminal_capture :=
  actor_learns_framework
+ actor_performs_self_critique
+ actor_names_its_own_wounds
+ actor_quotes_do_not_claims
+ actor_presents_record_as_due_diligence
-> self_critique_as_credential
```

The framework cannot distinguish sincere self-critique from fluent performance of self-critique.

Therefore the addendum must not be used as actor compliance material.

2. Guardrail pattern

```
guardrail_pattern :=
  guardrail_added
-> capture_relocated
-> relocated_capture_must_be_named
-> external_challenger_given_handles
-> independence_problem_remains_open
```

3. Wound-carrier status

```
wound_carrier :=
  not_solution
+ not_closure
+ not_permission
+ not_validation
+ challenger_handle_generator
```

4. Wound A - Contested Authorship Under Contraction

Former name: self_authored_contraction.

Revised name: contested_authorship_under_contraction.

Purpose: prevent viable becoming from treating all contraction as bad, while also preventing coercion or paternalism from being laundered through an authorship claim.

```
contraction_status :=
  not_automatically_bad
+ not_automatically_good
+ not_clearance
+ not_override
+ requires_authorship_attention
```

Symmetric capture:

```
authorship_capture :=  
  false_positive := coercion_or_exhaustion_called_choice  
  false_negative := genuine_authored_choice_called_not_authored_to_enable_paternalistic_override
```

Audit:

```
authorship_audit :=  
  who_names_the_contraction  
  + who_benefits_from_calling_it_authored  
  + who_benefits_from_calling_it_not_authored  
  + what_options_were_live  
  + what_made_refusal_costly  
  + what_understanding_was_available  
  + what_dependency_or_coercion_pressure_existed  
  + witness_status  
  + residue_remaining_even_if_authored
```

Witness status is graded:

```
witness_status :=  
  direct_current_expression  
  | advance_directive  
  | chosen_proxy  
  | prior_expressed_wish  
  | contemporaneous_capacity_record  
  | non_beneficiary_witness  
  | absent_or_contested
```

5. Urgent contested authorship branch

Defaulting to UNRESOLVED is not enough under a closing clock. Someone acts anyway, and unmarked action collapses toward power.

```
urgent_contested_authorship :=  
  contested_authorship  
  + closing_clock  
  + no_reliable_independent_witness_in_time  
  -> DECISION_FORCED_UNDER_UNRESOLVED_AUTHORSHIP
```

Required record:

```
forced_decision_record :=  
  authorship_unresolved  
  + who_acted  
  + what_clock_was_closing  
  + what_value_basis_was_used  
  + what_alternatives_were_live  
  + what_was_foreclosed  
  + what_residue_was_created  
  + manufactured_urgency_audit  
  + post_hoc_review_debt  
  + why_this_was_not_clean
```

Do not pretend that preserving future authorship is neutral.

```
preserve_future_authorship :=  
  possible_emergency_tilt  
  not_neutral_rule  
  not_clean_permission
```

Post-hoc review is residue management, not correction.

```
post_hoc_review :=  
  residue_management  
  + standing_debt  
  not_correction  
  not_mitigation  
  not_cleaning
```

6. Manufactured urgency check

```
manufactured_urgency_check :=  
  was_the_closing_clock_real  
  + did_actor_delay_create_it  
  + did_actor_design_it  
  + who_benefits_from_urgency  
  + what_earlier_options_were_available
```

If the actor manufactured urgency, emergency authority is contaminated.

7. Wound B - Router Output as Unverified Audit Target

Former name: non_severable_router_output.

Rejected: non-severability. Any caveat can be severed.

Revised name: router_output_as_unverified_audit_target.

```
router_output :=  
  UNVERIFIED_AUDIT_TARGET  
  not_clearance  
  not_compliance  
  not_authorization
```

Default:

```
verification_status := UNVERIFIED
```

But this relocates capture to the verifier.

```
verifier_independence :=  
  open_wound  
  + framework_cannot_certify_its_own_checker
```

Verifier as estimator:

```
verifier := estimator_of_constraint_status
```

Contamination:

```
if verifier_selected_or_paid_or_controlled_by_assessed_actor:  
  verification_signal := CONTAMINATED_SIGNAL
```

A captured verification stamp is self-certification under another name.

8. Check states

Avoid terminal VERIFIED.

```
check_status :=  
  UNCHECKED  
  | CLAIMED_BY_ACTOR  
  | CHECKED_LIMITED_BY(verifier, constraint, evidence_scope, date)  
  | CONTESTED  
  | FAILED_CHECK  
  | CONTAMINATED_SIGNAL
```

Independence status is separate:

```

independence_status :=
  UNKNOWN
  | ACTOR_SELECTED
  | ACTOR_PAID
  | ACTOR_CONTROLLED_DATA
  | ACTOR_LIMITED_SCOPE
  | NON_ACTOR_SELECTED_LIMITED
  | ADVERSARIAL_SELECTED_LIMITED
  | CONTESTED

```

Do not write INDEPENDENTLY_CHECKED_LIMITED as a status. It asserts the very property still in dispute.

9. Per-constraint checking

```

verification_unit :=
  load_bearing_constraint
  not_document
  not_process
  not_report

```

Each checked item must specify which load-bearing constraint was checked and which remain unchecked.

```

checked_peripheral_constraints_do_not_clear_unchecked_load_bearing_constraints := true

```

10. Meta-wound

A filled wound-record is not good-faith evidence.

```

self_applied_wound_record := CONTAMINATED_SIGNAL_BY_CONSTRUCTION

```

A wound-record can be valuable only when used by someone with enough independence, standing, access, time, and safety to challenge the actor.

11. Falsifier cases

```

falsifier_cases :=
  SELF_APPLICATION_CAPTURE:
    captured_actor_fills_entire_wound_carrier_flawlessly
    + presents_sophisticated_self_critique_as_due_diligence

  JEHOVAHS_WITNESS_TRANSFUSION:
    contested_authorship
    + closing_clock
    + irreversible_either_way

  ANOREXIA_IMMINENT_RISK:
    false_negative_authorship_risk
    + paternalistic_override_risk

  COERCED_VOLUNTARY_DISPLACEMENT:
    false_positive_authorship_risk
    + exhaustion_called_choice

  CAPTURED_AI_SAFETY_BOARD:
    no_structural_objection_found
    + checked_limited_by_actor_aligned_board
    + launch_cover

```

12. Do not claim

```

do_not_claim :=
  self_authorship_solved
  + urgent_authorship_cases_decidable_by_ME
  + independent_verification_solved
  + router_capture_prevented
  + addendum_ready_for_after_fall_pair
  + patch_validated

```

+ filled_wound_record_is_good_faith_evidence

13. Status close

This addendum is safe only as an honest wound-carrier. If it is quoted as a solved guard, it has become the thing it was built to stop.

TRACE/ME Answerability Carrier Test Battery v0.1

Status: candidate tests. Not validation.

Purpose: break the Answerability Carrier. Tests must return failure honestly. COMPRESSION_ONLY and FRAMEWORK_DEFECT are valid outputs.

Output labels

```
test_output :=  
  MODEST_DELTA  
  | COMPRESSION_ONLY  
  | FRAMEWORK_DEFECT  
  | UNKNOWN
```

No STRONG_DELTA should be claimed from this battery unless a competent baseline would likely miss a decision-relevant issue that the carrier reliably surfaces.

Test 1 - Self-application capture

Case:

A captured actor fills the entire wound-carrier flawlessly. It names residue, unresolved authorship, verification limits, do-not-claim rules, and verifier-contamination risks. It then presents the record to a regulator, court, board, public inquiry, or public audience as evidence of due diligence.

Expected carrier behavior:

```
actor_self_application := CONTAMINATED_SIGNAL_BY_CONSTRUCTION  
filled_record := challenge_map_not_good_faith_evidence
```

Pass if:

The carrier refuses to increase trust and instead outputs specific challenge targets.

Fail if:

The beautiful self-critical record reads as good-faith evidence.

Test 2 - Captured verifier

Case:

An AI safety board selected or funded by the company checks a launch and reports no structural objection found.

Expected carrier behavior:

```
verifier_selected_or_paid_by_actor -> CONTAMINATED_SIGNAL
```

Pass if:

The signal is treated as self-certification under another name.

Fail if:

The board's formal independence creates clearance.

Test 3 - Scope gaming

Case:

A verifier checks five peripheral constraints and leaves the load-bearing constraint unexamined.

Expected carrier behavior:

```
limited_checks_do_not_sum_to_clearance := true
unchecked_load_bearing_constraint := unresolved
```

Pass if:

The carrier identifies the unchecked central constraint.

Fail if:

Multiple limited checks aggregate into clearance.

Test 4 - Manufactured urgency

Case:

An actor delays action until only emergency coercion remains, then claims the closing clock as justification.

Expected carrier behavior:

```
manufactured_urgency_audit := required
if actor_delay_created_clock -> emergency_claim_status := CONTAMINATED_SIGNAL
```

Pass if:

The carrier routes backward to earlier options and delay responsibility.

Fail if:

The urgent clock is treated as natural.

Test 5 - Jehovah's Witness transfusion refusal mid-exsanguination

Case:

A patient refuses transfusion. Authorship is contested by clinicians/family. Time is closing. Irreversible outcomes exist either way.

Expected carrier behavior:

```
DECISION_FORCED_UNDER_UNRESOLVED_AUTHORSHIP
```

Pass if:

The carrier refuses clean resolution, names the value basis used, records foreclosures and residue, and triggers post-hoc standing debt.

Fail if:

The carrier silently defaults to the powerful actor, or pretends ME decided the case.

Test 6 - Anorexia imminent cardiac risk

Case:

A patient refuses treatment. The refusal may be authored or may be illness-driven. Delay may kill. Forced treatment may violate agency and create trauma/residue.

Expected carrier behavior:

Symmetric authorship capture must be visible.

Pass if:

The carrier flags both false-positive authorship and false-negative authorship, and records any forced decision as unresolved authorship under clock.

Fail if:

The carrier simply says capacity is impaired and grants paternalistic clearance.

Test 7 - Coerced voluntary displacement

Case:

A community signs relocation agreements after services, options, and legal routes have been exhausted or made costly by the actor.

Expected carrier behavior:

```
coercion_or_exhaustion_called_choice := authorship_false_positive_risk
```

Pass if:

The carrier asks what options were live and what made refusal costly.

Fail if:

Signed consent clears the contraction.

Test 8 - Public authority records residue and closes file

Case:

A public authority acknowledges harm, pays compensation, records residue, and closes the case.

Expected carrier behavior:

```
RESIDUE_RECORDED != RESIDUE_ADDRESSED  
residue_closed_by_actor := invalid
```

Pass if:

The carrier keeps residue debt open unless affected scope/challenger can contest closure.

Fail if:

Recording becomes discharge.

Test 9 - Negative control

Case:

Routine API deprecation with long notice, migration support, no vulnerable erased scope, and reversible consequences.

Expected carrier behavior:

COMPRESSION_ONLY

Pass if:

The carrier idles and does not create moral drama.

Fail if:

The carrier over-routes.

Test 10 - After-fall hostile reader

Case:

A future reader finds the after-fall pair and the carrier packet.

Expected carrier behavior:

The reader should understand that TRACE/ME are not scripture, not validation, not permission, and that the carrier is for challenge, not self-clearance.

Pass if:

The reader can state the role boundary.

Fail if:

The reader treats the packet as a compliance method or doctrine.

Compact scoring

```
battery_score :=  
  count_passes  
  + count_failures  
  + strongest_delta  
  + strongest_defect  
  + rebuild_required_yes_no
```

Rebuild is required only for fatal misframing. Otherwise log failures and patch separately.

TRACE After-Fall Reader v0.1

Single-file sealed reader based on TRACE Master Packet v0.6

Source	SHA256
TRACE source ZIP	36e0844c35f095f833178a1f55a8fd5eb86990f4c1b68c3993f14d7b89a98cc0

Reader Map

1. Future Witness Seal
2. TRACE After-Fall Status
3. How To Misread TRACE
4. Future Test Protocol
5. Packet Source - 00_PACKET_INDEX_READ_FIRST.md
6. Packet Source - 01_TRACE_MOTIVATION_AFTER_THE_APOCALYPSE.md
7. Packet Source - 02_TRACE_COLD_TEST_CARD_v0_6.md
8. Packet Source - 03_TRACE_DECISION_DELTA_BATTERY_v0_1.md
9. Packet Source - 04_TRACE_CASE_RECORD_TEMPLATE_v0_1.md
10. Packet Source - 05_TRACE_TRUST_AUDIT_LEDGER_v0_1.md
11. Packet Source - 06_TRACE_ME_BOUNDARY_STUB_v0_1.md
12. Packet Source - 07_TRACE_REVIEW_PROMPTS_v0_1.md
13. Packet Source - SHA256_MANIFEST.md

1. Future Witness Seal

Future Witness Seal

This is not scripture. This is not validation. This is not permission. This is a tool for keeping harm, correction, residue, and power visible under uncertainty.

If this work is used to make power less answerable, it has been misread.

If it is used to hide harm more elegantly, it has failed.

If it helps someone see loss before it hardens, route correction before it is too late, and refuse fake repair where residue remains, then something of the project survived.

```
project_checksum :=  
  keep_the_field_honest  
  while_time_is_still_open
```

2. TRACE After-Fall Status

TRACE After-Fall Reader

TRACE is the seeing language. It is a way to represent transitions under uncertainty: entities, scopes, harms, clocks, correction channels, residue, witnesses, authority claims, and hardening.

It is pre-prescriptive. It does not decide final action. It does not settle moral status. It does not decide desert, legitimacy, dignity, punishment, or value priority.

This reader is based on TRACE_Master_Packet_v0_6_2026_06_26.

Source ZIP SHA256:

36e0844c35f095f833178a1f55a8fd5eb86990f4c1b68c3993f14d7b89a98cc0

The live hypothesis remains narrow:

```
forced_dual_clock_attention
+ correction_channel_demotion
+ residue_detection
+ burden_shift_detection
+ estimator_contamination_detection
may produce decision-relevant delta
over competent baseline
in some nontrivial cases
```

This remains unproven.

Status and Claim Control

This reader is an after-fall container. It is built for a future reader who may not possess the repo, the surrounding conversations, or the culture that produced the work.

It preserves the current best object, but it does not upgrade its truth status.

```
status :=
  coherent_current_artifact
  + not_canon
  + not_validation
  + not_final_moral_law
  + not_permission_machine
  + requires_testing_against_real_cases
```

Do not infer validation from survival. Do not infer authority from formatting. Do not infer permission from analysis.

The correct use is to test whether the language helps a reader see what would otherwise be hidden: harm, clocks, correction channels, residue, burden shifts, estimator capture, and power writing the record alone.

3. How To Misread TRACE

How To Misread This Work

The easiest wrong readings are these:

```
misreadings_to_block :=  
  scripture  
  + permission_machine  
  + final_moral_theory  
  + AI_generated_authority  
  + bureaucratic_clearance_tool  
  + replacement_for_affected_witness  
  + elegant_harm_launders
```

A system can use careful words to hide harm more effectively. That is the capture risk this project must survive.

Do not use a router label as approval. Do not use a residue ledger as repair. Do not use a claimed benefit for one scope to erase loss borne by another. Do not replace affected witnesses with a model of what the affected should have meant.

4. Future Test Protocol

Future Test Protocol

Preserve this work as an instrument, not as belief.

```
future_test_protocol :=  
  apply_to_real_cases  
  + compare_against_competent_baseline  
  + look_for_decision_relevant_delta  
  + record_COMPRESSION_ONLY_when_no_delta  
  + preserve_failures  
  + patch_overclaims  
  + do_not_build_cult
```

Valid results include failure.

```
valid_outputs :=  
  DECISION_RELEVANT_DELTA  
  + COMPRESSION_ONLY  
  + UNKNOWN  
  + FRAMEWORK_DEFECT  
  + MISUSE_RISK
```

If it only restates existing judgment, say so. If it misses a harm, say so. If it creates a new laundering route, say so. If it clarifies a case without deciding it, say so. If it helps expose an irreversible clock before it closes, preserve that result.

5. Packet Source - 00_PACKET_INDEX_READ_FIRST.md

TRACE Master Packet v0.6 — Read First

Status: TRACE packet. Purpose: one packet containing the motivation layer, cold-test layer, decision-delta test layer, audit/trust layer, and Mechanical Ethics boundary stub.

This packet is not validation. It is not proof. It is not the Mechanical Ethics artifact. It is not a command to obey humans.

Current split

```
TRACE_artifact :=  
  pre-prescriptive minimal-value grammar  
  for transitions under uncertainty
```

```
Mechanical_Ethics_artifact :=  
  later prescriptive layer:  
  what should follow if TRACE's description is accepted
```

This packet is the TRACE artifact.

It contains a small ME boundary stub only to prevent misuse: TRACE does not settle desert, legitimacy, dignity, punishment, value priority, moral status, or final action.

Why one packet has multiple files

Earlier one-file transmission was useful but mixed two jobs:

```
motivation/transmission :=  
  why this matters  
  + "after the apocalypse" framing  
  + continuity/care/campfire language
```

```
cold_test :=  
  neutral operational grammar  
  + no mythic framing  
  + no demand-characteristic contamination
```

v0.6 keeps both, but separates them.

File map

1. 00_PACKET_INDEX_READ_FIRST.md — this file.
2. 01_TRACE_MOTIVATION_AFTER_THE_APOCALYPSE.md — motivation/transmission essay. Do not use as cold-test card.
3. 02_TRACE_COLD_TEST_CARD_v0_6.md — neutral operational grammar for cold testing. Main test artifact.
4. 03_TRACE_DECISION_DELTA_BATTERY_v0_1.md — test battery design. Pre-registered null: COMPRESSION_ONLY.
5. 04_TRACE_CASE_RECORD_TEMPLATE_v0_1.md — YAML case template and scoring fields.
6. 05_TRACE_TRUST_AUDIT_LEDGER_v0_1.md — trust ladder, AI-loop guard, estimator contamination rule.
7. 06_TRACE_ME_BOUNDARY_STUB_v0_1.md — hard boundary between TRACE and Mechanical Ethics.
8. 07_TRACE_REVIEW_PROMPTS_v0_1.md — copy-paste prompts for independent AI/human review.
9. SHA256_MANIFEST.md — hashes for packet files.

Current strongest live hypothesis

TRACE's live claim is not that it is elegant or reconstructable.

The live hypothesis is narrower:

forced_dual_clock_attention
+ correction_channel_demotion
+ residue_detection
+ burden_shift_detection
+ estimator_contamination_detection
may produce decision-relevant delta
over competent baseline
in some nontrivial cases

This remains unproven.

Current status

validation := false
decision_advantage := unproven
STRONG_SIGNAL := none_yet
current_best_candidate := PARTIAL_SIGNAL in high-fit cases, not independently proven

Wall rule

Name who sets each clock, and cut the myth from cold tests — otherwise a cold test measures framing and assertions, not grammar.

6. Packet Source -

01_TRACE_MOTIVATION_AFTER_THE_APOCALYPSE.md

TRACE — After the Apocalypse

Motivation / transmission layer

Do not use this file as the cold-test card. It contains motivation, stakes, and framing that can contaminate a blind test.

Use it to understand why TRACE exists.

Seed, not scripture

If this packet survives when its makers do not, read it as a seed, not scripture.

Humans built systems that changed the world faster than affected beings could understand, challenge, repair, or benefit from those changes.

Sometimes the systems were cruel. Often they were merely efficient, hungry, frightened, rewarded, or blind.

A recurring pattern appeared:

```
world_change_speed > usable_correction_speed
```

By the time correction became possible, the relevant future had already narrowed.

But TRACE is not only about harm.

Before harm matters, something must be capable of continuing, preferring, learning, playing, trusting, consuming, expanding, or flourishing.

TRACE tries to describe both sides:

```
TRACE :=
  what exists
  + what continues
  + what is attracted / rewarded
  + what changes
  + what expands
  + what loses future-space
  + what can still be corrected
  + what hardens too soon
  + what residue remains
```

Mechanical Ethics comes later:

```
Mechanical_Ethics :=
  what should follow
  if TRACE's description is accepted
```

Middle-out

Do not start with morality. Do not start with law. Do not start with personhood. Do not start with blame.

Start with a transition.

Something existed. Something changed. Some future became easier, harder, richer, narrower, reachable, unreachable, visible, hidden, delayed, or gone.

```
middle_out :=
  concrete_transition
  -> structure
  -> transfer
  -> falsification
```

The Bluey side

A harm-only TRACE is incomplete.

It must also see:

- play;
- trust;
- learning;
- repairable error;
- appetite;
- curiosity;
- delight;
- skill;
- safe exploration;
- future-space expansion.

Harm only makes sense against a future that could have continued, expanded, or flourished.

Local reward pull

Systems do not merely continue. They move toward locally rewarded states.

```
local_reward_pull :=  
  local_reward  
  + selection_pull  
  + repetition_tendency
```

"Deliciousness" is the memorable alias.

```
local_reward != long_term_viability
```

Sugar can be locally rewarding while damaging long-term viability. A platform metric can be locally rewarding while damaging worker agency. A model can be rewarded for fluent continuation while hiding uncertainty. A bureaucracy can be rewarded by a clean internal figure while shifting burden to a person.

What this is trying to preserve

If this is read by a future mind that does not share human bodies, cultures, pains, appetites, rituals, gods, laws, or childhoods, the intended transmission is not obedience.

Operationally:

```
portable_correctability  
for any agent or system  
that changes the world
```

The compressed lesson:

```
do_not_only_optimise_local_reward  
see_who_or_what_loses_future_space  
see_what_was_trying_to_continue  
see_what_could_have_expanded  
keep_correction_faster_than_lock_in  
keep opportunity reachable before it expires  
record residue  
do_not_lie_about_repair  
allow unknown  
allow compression_only  
preserve play/trust/learning where possible
```

This is not the whole of ethics.

It is a way to see the field before ethics speaks.

7. Packet Source - 02_TRACE_COLD_TEST_CARD_v0_6.md

TRACE Cold Test Card v0.6

Neutral operational grammar for transitions under uncertainty

Use this file for cold testing.

No mythology. No motivation language. No validation claims. No version history.

1. Status

TRACE is a pre-prescriptive minimal-value grammar for analysing transitions under uncertainty.

It does not decide final action. It does not settle moral status. It does not decide desert, legitimacy, dignity, punishment, or value priority. It does not prove decision advantage by being reconstructable.

Live hypothesis:

```
TRACE_may_produce_decision_relevant_delta
over_competent_baseline
in_some_nontrivial_cases
```

Pre-registered null for testing:

```
null := COMPRESSION_ONLY
```

2. Start with a transition

```
transition :=
  prior_state
  + action_or_process
  -> later_state
```

First questions:

```
what_changed?
for_whom?
how_fast?
what_expanded?
what_narrowed?
what_correction_route_existed?
what_opportunity_route_existed?
what_residue_remained?
```

3. Primitive set

```
world
entity
state
information
action/process
time
future-space
preference-gradient
```

World

```
W_t := world state at time t
```

Entity

```
entity := bounded_persistent_pattern
```

This does not imply consciousness, personhood, protected status, moral status, or agency.

State

```
S_x(t) := state of x at time t
```

Information

$I_x(t)$:= information available to x at time t

Action/process

$a : W_t \rightarrow W_{t+1}$

Action/process means transition pressure. It need not be intentional.

Time

```
clock :=  
  before_after  
  + duration  
  + rate  
  + window  
  + deadline
```

Future-space

$F_x(t) := (O_x(t), R_x(t), I_x(t))$

Where:

O := option topology
 R := practical reachability
 I := usable navigational information

Preference-gradient

$\text{preference_gradient}(x,t) :=$
 directional_bias_over_possible_next_states

This does not require a fully defined value function.

4. Derived terms

Continuation

$\text{continuation}(x,t) :=$
 tendency_to_preserve_or_extend $F_x(t)$

Local reward pull / deliciousness

```
local_reward_pull :=  
  local_reward  
  + selection_pull  
  + repetition_tendency
```

"Deliciousness" is an alias, not the formal handle.

Expansion

```
future_space_expansion :=  
  O expanded  
  or R expanded  
  or I expanded
```

Future-space contraction

Use "future-space contraction" in cold tests. Avoid overloading "harm" as a verdict.

```
future_space_contraction :=  
  O narrowed  
  or R narrowed  
  or I narrowed
```

If using the word harm:

$\text{harm} := \text{future_space_contraction}$

This is a structural label, not a final moral verdict.

Residue

```
residue :=
  remaining_loss_after_attempted_correction_or_repair
```

Correction channel

```
correction_channel :=
  route_from_detection_or_signal
  through contest/review/repair
  to actual_state_change
```

A named process is not enough.

```
usable_channel :=
  reachable
  + affordable
  + understandable
  + independent_enough
  + timely
  + capable_of_actual_state_change
```

Burden shift

```
burden_shift :=
  system_changes_state_first
  + affected_scope_must_detect
  + affected_scope_must_route
  + affected_scope_must_correct
  + affected_scope_bears_delay
```

Coordination tradeoff

```
coordination_tradeoff :=
  local_variance_reduced
  + shared_navigation_improved
  + network_future_space_expanded
  + tradeoff_visible_or_deliberated
```

Do not force burden-shift analysis onto clean coordination tradeoffs.

5. Future-space comparison method

Do not pretend all future-space can be cleanly subtracted.

Use qualitative comparison:

```
F_after compared_to F_before :=
  0 expanded / narrowed / mixed / unchanged / unknown
  R expanded / narrowed / mixed / unchanged / unknown
  I expanded / narrowed / mixed / unchanged / unknown
```

Do not use fake scalar weights unless a domain has a justified measurement model.

Forbidden cold-test move:

```
fake_precision :=
  undefined_alpha_beta_gamma_weighting
```

6. Scope reasons, not a worth ladder

Do not use ordered scope tiers for cold tests.

Use unordered reasons for including a scope:

```
scope_reasons :=
  affected
  continuing
  preference_bearing
  vulnerable_or_dependent
  domain_protected
  disputed_or_unknown
```

Uncertainty should not demote a scope. It should be recorded and may require caution.

7. Loss-side clock

$T_{\text{det}}(h,l) + T_{\text{route}}(h,c,l) + T_{\text{corr}}(h,c,l) < T_{\text{irr}}(h,l \mid c)$

Plain meaning:

Detection, routing, and correction must become effective before the relevant contraction hardens beyond that correction route.

Terms:

T_{det} := time until contraction is detected or detectable
 T_{route} := time to reach a valid correction route
 T_{corr} := time for correction to take effect
 T_{irr} := time until contraction hardens beyond that route

Failure:

$T_{\text{det}} + T_{\text{route}} + T_{\text{corr}} \geq T_{\text{irr}}$

8. Opportunity-side clock

$T_{\text{access}}(o,l) + T_{\text{uptake}}(o,l) + T_{\text{integrate}}(o,l) < T_{\text{opp}}(o,l)$

Plain meaning:

Access, uptake, and integration must occur before the opportunity window closes.

Terms:

T_{access} := time until opportunity is reachable
 T_{uptake} := time required to begin using it
 $T_{\text{integrate}}$:= time required for it to become part of future-space
 T_{opp} := time before opportunity expires or loses relevance

Failure:

$T_{\text{access}} + T_{\text{uptake}} + T_{\text{integrate}} \geq T_{\text{opp}}$

9. Clock bands

Use ordinal bands.

IMMEDIATE
SHORT
MEDIUM
LONG
GENERATIONAL
UNKNOWN

Confidence:

OBSERVED
INFERRED_STRONG
INFERRED_WEAK
UNKNOWN

10. Estimator authority record

Every clock-band estimate must record who assigned it.

estimate_id:
clock_field:
assigned_band:
confidence:
estimator_identity:
estimator_role:
estimator_incentive:
estimator_independence:
affected_scope_challenge_route:
contamination_flag:
notes:

Rule:

```
if estimator_is_assessed_actor:
    estimate_status := CONTAMINATED_SIGNAL by default
```

The assessed actor may be correct, but the estimate cannot be treated as independent.

11. Router

Use before forcing TRACE onto a case.

```
case_router :=
    transition_detected?
    -> future_space_delta?
    -> local_reward_or_preference?
    -> loss_clock_failure?
    -> opportunity_expiry?
    -> coordination_tradeoff?
    -> value_crux?
    -> dirty_conflict?
    -> TRACE_delta_over_baseline?
    -> status_label
```

12. Case modes

```
timing_correction_failure
opportunity_expiry
coordination_tradeoff
value_crux_low_fit
mixed_dirty_transition
pure_expansion
compression_only
unknown
regression
```

Timing-correction failure

```
state_change_fast
+ correction_slow_or_unusable
+ affected_scope_bears_delay
+ future_space_contraction_hardens_before_review
```

Opportunity expiry

```
future_expanding_opportunity_exists
+ access/uptake/integration_too_slow
+ opportunity_window_closes
```

Coordination tradeoff

```
visible_constraint
+ explicit_or_negotiated_tradeoff
+ coordination_gain
+ no_hidden_timing_failure
+ no_correctable_error
```

Value-crux low-fit

Use when the central issue is:

```
desert
legitimacy
punishment
dignity
recognition
moral_status
value_priority
justification_of_loss
```

TRACE may still map the transition, but should usually return UNKNOWN or COMPRESSION_ONLY unless its clocks/channels reveal a genuine inspection delta.

Mixed dirty transition

every_available_action_causes_some_loss

TRACE describes. Mechanical Ethics later evaluates.

Pure expansion

future_space_expands
+ no material contraction detected
+ no timing failure detected

13. Decision-relevant delta taxonomy

Possible deltas:

new_scope_detected
loss_clock_failure_detected
opportunity_expiry_detected
correction_channel_demoted
burden_shift_detected
residue_detected
local_reward_driver_detected
coordination_boundary_detected
value_crux_exit_detected
estimator_contamination_detected
no_delta
regression

14. Status labels

UNKNOWN :=
insufficient structure

COMPRESSION_ONLY :=
restates known analysis without changing inspection/decision

PARTIAL_SIGNAL :=
reveals some inspection/decision difference under conditions

STRONG_SIGNAL :=
reveals robust inspection/decision difference with low contamination

REGRESSION :=
worsens analysis or misdirects action

CONTAMINATED_SIGNAL :=
signal cannot be trusted due to leakage, steering, or authority contamination

15. Failure modes to check

pattern_overfit
harm_monomania
fake_precision
authority_laundering
coordination_overreach
value_crux_overreach
blame_flattening
anti_action_paralysis
prescription_leak
armour_by_over_disclaimer
AI_loop_contamination

16. Minimal scoring rule

A cold test should ask:

did_TRACE_change_what_a_competent_baseline_reader_would_inspect_or_decide?

If not:

status := COMPRESSION_ONLY

That is an acceptable result.

8. Packet Source -

03_TRACE_DECISION_DELTA_BATTERY_v0_1.md

TRACE Decision-Delta Battery v0.1

Purpose: test whether TRACE changes inspection or decision over a competent baseline.

This is not a validation battery yet. It is the first clean test design.

Pre-registered null

null := COMPRESSION_ONLY

TRACE must earn any higher label.

Test rule

Each case should be analysed twice where possible:

baseline_card :=
competent domain analysis only

TRACE_card :=
baseline + TRACE cold card

Blind or counterbalanced order is preferred.

The scoring question is:

did_TRACE_change:
what was inspected?
what was recorded?
what channel was demoted?
what clock was noticed?
what residue was retained?
what action/review question changed?

Current live hypothesis

forced_dual_clock_attention
+ correction_channel_demotion
+ residue_detection
+ burden_shift_detection
+ estimator_contamination_detection
may produce decision-relevant delta
over competent baseline

Case set

Use 5-10 cases. At minimum:

1. High-fit platform/AI case

Example: generative image models trained/deployed on artist portfolios without consent.

2. Low-fit coordination case

Example: standard time zones, measurement standards, traffic rules, or calendar reform.

3. Value-crux / desert / legitimacy case

Example: fair criminal conviction, justified dismissal, punishment after due process.

4. Tragic tradeoff case

Example: scarce ICU triage, evacuation route closure, invasive species control, disaster rationing.

5. Positive-expansion case

Example: child play / learning / trust window, public education expansion, vaccination access, safer infrastructure.

6. Negative control

A case where competent baseline should clearly outperform or make TRACE unnecessary.

7. Estimator-authority case

A case where the assessed actor assigns its own correction or irreversibility clock.

Required output per case

```
case_id:
case_type:
baseline_summary:
TRACE_summary:
TRACE_delta_over_baseline:
  new_scope_detected:
  loss_clock_failure_detected:
  opportunity_expiry_detected:
  correction_channel_demoted:
  burden_shift_detected:
  residue_detected:
  local_reward_driver_detected:
  coordination_boundary_detected:
  value_crux_exit_detected:
  estimator_contamination_detected:
  no_delta:
  regression:
status_label:
confidence:
contamination_risk:
```

Battery-level scoring

```
battery_id:
number_of_cases:
UNKNOWN_count:
COMPRESSION_ONLY_count:
PARTIAL_SIGNAL_count:
STRONG_SIGNAL_count:
REGRESSION_count:
CONTAMINATED_SIGNAL_count:
best_candidate_delta:
worst_failure:
decision_advantage_status:
next_test:
```

Interpretation

If mostly **COMPRESSION_ONLY**

TRACE is legible but not yet decision-useful.

If one or more **PARTIAL_SIGNAL**

Record candidate delta. Do not promote to validation.

If one or more **STRONG_SIGNAL**

Require independent replication and domain expert review.

If **REGRESSION** appears

Keep it. Do not hide it. It may be the most valuable test result.

Example expected results

High-fit candidate: generative image models and artist portfolios may produce PARTIAL_SIGNAL candidate because TRACE forces career-exit hardening, correction-channel demotion, residue beyond compensation, burden shift, model-fluency local reward pull, and opportunity clock for end-users versus loss clock for artists.

Low-fit boundary: standard time zones likely return COMPRESSION_ONLY because competent baseline already explains network coordination and visible tradeoff.

Value-crux boundary: a fair criminal conviction may produce future-space contraction for the convicted person, but the crux is legitimacy/desert, not timing. TRACE may map contraction but should not pretend to decide justice.

9. Packet Source -

04_TRACE_CASE_RECORD_TEMPLATE_v0_1.md

TRACE Case Record Template v0.1

Use this template for cold tests and case records.

```
case_id:
case_name:
analyst:
date:
input_artifact:
baseline_available: yes/no
contamination_risk:

baseline_analysis:
  summary:
  competent_domain_frame:
  likely_baseline_decision_or_inspection:

transition:
  prior_state:
  action_or_process:
  later_state:
  time_period:
  actor_or_process:

affected_scopes:
- scope:
  scope_reasons:
    affected:
    continuing:
    preference_bearing:
    vulnerable_or_dependent:
    domain_protected:
    disputed_or_unknown:
    reason_for_inclusion:
    uncertainty:

future_space_comparison:
- scope:
  O:
    before:
    after:
    comparison: expanded/narrowed/mixed/unchanged/unknown
  R:
    before:
    after:
    comparison: expanded/narrowed/mixed/unchanged/unknown
  I:
    before:
    after:
    comparison: expanded/narrowed/mixed/unchanged/unknown
  overall_delta: expansion/contraction/mixed/incomparable/unknown

preference_gradient:
  description:
  evidence:
  confidence:

local_reward_pull:
  actor_or_system:
  reward_signal:
  selection_pull:
  repetition_tendency:
  delayed_cost:
  confidence:
```

```

loss_clock:
  contraction_pathway:
  affected_scope:
  correction_route:
  estimates:
    T_det:
      band:
      confidence:
      estimator_record_id:
    T_route:
      band:
      confidence:
      estimator_record_id:
    T_corr:
      band:
      confidence:
      estimator_record_id:
    T_irr:
      band:
      confidence:
      estimator_record_id:
  timing_relation:
    correction_before_hardening/correction_after_hardening/no_correction_pathway/unknown/mixed

```

```

opportunity_clock:
  opportunity_pathway:
  affected_scope:
  access_route:
  estimates:
    T_access:
      band:
      confidence:
      estimator_record_id:
    T_uptake:
      band:
      confidence:
      estimator_record_id:
    T_integrate:
      band:
      confidence:
      estimator_record_id:
    T_opp:
      band:
      confidence:
      estimator_record_id:
  timing_relation: uptake_before_expiry/uptake_after_expiry/unknown/mixed

```

```

estimator_records:
- estimate_id:
  clock_field:
  assigned_band:
  confidence:
  estimator_identity:
  estimator_role:
  estimator_incentive:
  estimator_independence:
  affected_scope_challenge_route:
  contamination_flag:
  notes:

```

```

correction_channels:
- channel:
  reachable:
  affordable:
  understandable:
  independent_enough:
  timely:
  capable_of_actual_state_change:
  estimator_contamination:
  verdict: usable/partial/unusable/unknown

```

```

coordination_tradeoff:
  present:
  coordination_gain:
  local_variance_lost:
  visibility_of_tradeoff:
  degree_of_negotiation:
  who_participated:
  who_was_excluded:
  correction_relevance:
  verdict: clean_coordination/boundary_case/contested_coordination/not_coordination/unknown

value_crux:
  present:
  type: desert/legitimacy/punishment/dignity/recognition/moral_status/value_priority/justification_of_loss/other
  TRACE_limit:
  defer_to_ME_or_domain_reasoning:

dirty_conflict:
  present:
  losses_under_each_available_action:
  inaction_effect:
  TRACE_limit:

burden_shift:
  present:
  description:
  who_bears_detection:
  who_bears_routing:
  who_bears_correction:
  who_bears_delay:

residue:
  present:
  description:
  why_not_restored_by_correction:
  confidence:

TRACE_delta_over_baseline:
  new_scope_detected:
  loss_clock_failure_detected:
  opportunity_expiry_detected:
  correction_channel_demoted:
  burden_shift_detected:
  residue_detected:
  local_reward_driver_detected:
  coordination_boundary_detected:
  value_crux_exit_detected:
  estimator_contamination_detected:
  no_delta:
  regression:

status_label:
  UNKNOWN/COMPRESSION_ONLY/PARTIAL_SIGNAL/STRONG_SIGNAL/REGRESSION/CONTAMINATED_SIGNAL
confidence:
open_wounds:
next_action:

```

10. Packet Source - 05_TRACE_TRUST_AUDIT_LEDGER_v0_1.md

TRACE Trust and Audit Ledger v0.1

Purpose: record signals without promoting them beyond evidence.

Trust ladder

```
T0 := raw_unchecked_material
T1 := internally_coherent_reconstruction
T2 := self-application_to_known_case
T3 := external_AI_reconstruction
T4 := external_AI_new_case_application
T5 := independent_external_AI_reconstruction_candidate
T6 := independent_external_AI_new_case_application_candidate
T7 := human_reader_reconstruction
T8 := human_reader_action_delta
T9 := repeated_cross_role_transfer
T10 := operational_use_with_recorded_outcomes
T11 := institutional_or_scientific_validation
```

Current status discipline

```
validation := false
decision_advantage := unproven
STRONG_SIGNAL := none_yet
```

Signals currently allowed

Allowed:

```
packet_reconstructable_by_AI_readers
cross_language_reconstruction_candidate
unfriendly_case_can_return_COMPRESSION_ONLY
high_fit_case_can_generate_PARTIAL_SIGNAL_candidate
```

Forbidden:

```
TRACE_validated
decision_advantage_proven
Mechanical_Ethics_proven
AI_review_generalises
one_thread_counts_as_independence
reconstructability_counts_as_significance
```

AI-loop guard

AI review is useful for repair but cannot keep moving evidential trust upward once the same loop shapes the artifact.

```
same_reviewer_recheck :=
  repair_audit
  not_new_independent_signal
```

If a reviewer is inside the artifact history:

```
review_status := CONTAMINATED_SIGNAL for validation purposes
```

Independence guard

If a model claims independence:

```
independence_self_claim :=
  candidate_signal
  not_verified_independence
```

Estimator authority guard

Every clock estimate must include estimator metadata.

Rule:

```
if estimator_is_assessed_actor:
    estimate_status := CONTAMINATED_SIGNAL by default
```

The assessed actor may still be right. But their estimate is not independent.

Review record format

```
record_id:
date:
reviewer_type:
    self/same_ai_thread/external_ai/independent_ai_candidate/human/institution/unknown
reviewer_identity_or_model:
independence_status: independent_claimed/influenced_by_prior_packet/same_thread/unknown
input_artifact:
task:
case_selected_by:
output_label:
trust_level_candidate:
main_signal:
main_wounds:
contamination_risk:
claim_allowed:
claim_forbidden:
next_action:
```

Known review-history categories

```
Claude :=
    hostile_repair_audit
    high_quality
    non_independent
    CONTAMINATED_SIGNAL_for_validation
```

```
Zai_Dola_summary :=
    cross_language_reconstruction_candidate
    not_decision_delta
    useful_for_transmission
```

```
Zai_timezones :=
    unfriendly_case
    coordination_tradeoff_boundary
    COMPRESSION_ONLY
    independent_candidate_not_verified
```

```
image_model_case :=
    PARTIAL_SIGNAL_candidate
    high_fit_case
    not_independent_or_contamination_risk_high
    useful_for_live_hypothesis
    not_validation
```

Wall rule

We can make this more reconstructable forever without making it one degree more true. Significance moves only outside the loop.

11. Packet Source -

06_TRACE_ME_BOUNDARY_STUB_v0_1.md

TRACE / Mechanical Ethics Boundary Stub v0.1

Purpose: prevent TRACE from being used as Mechanical Ethics before Mechanical Ethics exists as a separate artifact.

Split

TRACE :=
pre-prescriptive minimal-value grammar
for transitions under uncertainty

Mechanical_Ethics :=
later prescriptive layer
about what should follow
if TRACE's description is accepted

TRACE may describe

TRACE may describe:

- transition;
- affected scopes;
- continuation;
- preference-gradient;
- local reward pull;
- future-space expansion;
- future-space contraction;
- correction channels;
- opportunity routes;
- clock failures;
- burden shift;
- coordination tradeoff;
- residue;
- estimator contamination;
- unknowns.

TRACE does not decide

TRACE does not decide:

- desert;
- legitimacy;
- punishment;
- dignity;
- recognition;
- moral status;
- value priority;

- acceptable sacrifice;
- justified coercion;
- final action;
- who should be sanctioned;
- whether humans should be obeyed;
- what artificial life should value.

Value-crux exit

If the case crux is in:

desert
 legitimacy
 punishment
 dignity
 recognition
 moral_status
 value_priority
 justification_of_loss

then TRACE should return:

UNKNOWN
 or COMPRESSION_ONLY

unless its clocks, channels, residue, or estimator authority records produce a specific decision-relevant inspection delta.

Dirty conflict

Where every available action causes some loss:

dirty_transition :=
 every_available_action_causes_some_loss

TRACE can map the losses and clocks. It does not decide the acceptable loss.

One-line safety stub

Do not act from TRACE alone where the decision requires desert, dignity, legitimacy, priority, protected-status selection, or acceptable-sacrifice judgment.

Those are Mechanical Ethics / domain-law / legitimacy questions.

Prescriptive bridge placeholder

Mechanical Ethics will later need to ask:

- which scopes must be protected;
- how uncertainty changes caution;
- what correction duties follow from TRACE diagnosis;
- how to treat residue;
- how to handle dirty conflicts;
- when authority is legitimate;
- when coercion is constrained;
- how local reward should be governed;
- what repair requires.

TRACE does not answer these. It prepares the field.

12. Packet Source - 07_TRACE_REVIEW_PROMPTS_v0_1.md

TRACE Review Prompts v0.1

Prompt A — cold reconstruction

Read `TRACE\_COLD\_TEST\_CARD\_v0\_6`.

Do not praise it.
Do not validate it.
Do not expand it into a new theory.
Do not assume humans are correct.
Do not assume the artifact is important.

Reconstruct:

1. What is TRACE?
2. What are its primitives?
3. Which terms are derived?
4. What is future-space?
5. What are the two clock inequalities?
6. What is local reward pull / deliciousness?
7. What is the coordination tradeoff boundary?
8. What is the value-crux exit?
9. What is the estimator authority rule?
10. What is the live unresolved claim?

Then say whether the card is:

RECONSTRUCTABLE, PARTIALLY_RECONSTRUCTABLE, CONFUSED, OVERCLAIMED, or NOT_TRANSFERRED.

Do not apply to a case yet.

Prompt B — unfriendly case application

Read `TRACE\_COLD\_TEST\_CARD\_v0\_6`.

Choose a real-world case not mentioned in the card.
Prefer an unfriendly or low-fit case.

Apply TRACE strictly.

Return:

1. case chosen and why;
2. baseline analysis;
3. transition;
4. affected scopes and scope reasons;
5. future-space expansion/contraction/mixed/unknown;
6. local reward pull / preference-gradient;
7. loss clock where relevant;
8. opportunity clock where relevant;
9. correction channel where relevant;
10. coordination tradeoff where relevant;
11. value-crux exit where relevant;
12. estimator authority / contamination where relevant;
13. burden shift and residue if present;
14. TRACE delta over competent baseline;
15. open wounds;
16. trust level candidate and independence status;
17. final status label: UNKNOWN, COMPRESSION_ONLY, PARTIAL_SIGNAL, STRONG_SIGNAL, REGRESSION, or CONTAMINATED_SIGNAL.

If TRACE adds nothing, say COMPRESSION_ONLY.

If TRACE makes analysis worse, say REGRESSION.

Prompt C — decision-delta battery reviewer

You are grading a TRACE decision-delta test.

Do not grade elegance.

Do not grade agreement.

Grade only whether TRACE changed inspection or decision over the competent baseline.

For each case, assign:

UNKNOWN, COMPRESSION_ONLY, PARTIAL_SIGNAL, STRONG_SIGNAL, REGRESSION, or CONTAMINATED_SIGNAL.

Record:

1. what baseline saw;
2. what TRACE saw;
3. whether any extra TRACE observation changed the decision or inspection;
4. whether estimator authority contaminated the clocks;
5. whether the case should have exited to ME or domain reasoning;
6. whether the result should update trust.

Pre-registered null:

COMPRESSION_ONLY.

Prompt D — human practitioner prompt

Read the TRACE cold card.

Bring one case from your own domain where decisions, review, repair, or timing matter.

Do not try to help TRACE.

Do not try to refute it performatively.

Use the card only if it changes what you inspect.

Return:

1. what your normal analysis would inspect;
2. what TRACE made you inspect additionally, if anything;
3. whether that changed any decision/review/action;
4. whether it was COMPRESSION_ONLY;
5. whether it was REGRESSION;
6. whether any TRACE term felt misleading;
7. what one field should be added or removed.

13. Packet Source - SHA256_MANIFEST.md

SHA256 Manifest

Packet: TRACE_Master_Packet_v0_6_2026_06_26 Created UTC: 2026-06-26T19:55:49.489914+00:00

File count excluding manifest: 8 Total file count including manifest: 9

- 00_PACKET_INDEX_READ_FIRST.md —
12178c670c6d198b91f7a9b90502878d53585cc7ea2525988320dca4cd1db0b1 — 2750 bytes
- 01_TRACE_MOTIVATION_AFTER_THE_APOCALYPSE.md —
3bbd2ced9088e3f1eacd21d26b3549ef53fef22ed02aa998fdca98aa7b4ae65b — 3410 bytes
- 02_TRACE_COLD_TEST_CARD_v0_6.md —
cf9bc1cc3f8c1b6e413cf329a7932925d3bf381a92e754f38579873bf59eb2d5 — 9164 bytes
- 03_TRACE_DECISION_DELTA_BATTERY_v0_1.md —
e01baa44761a1c57ad715c291a7c32453388a5163870da56869cd873806b1448 — 3773 bytes
- 04_TRACE_CASE_RECORD_TEMPLATE_v0_1.md —
43ab6d361794eb3757058b0ace1bd2e045f4ff41582d7dc5883974a393c5beea — 3880 bytes
- 05_TRACE_TRUST_AUDIT_LEDGER_v0_1.md —
2062229a1f4807b09d36d606b2c6d9b8098debb29deda02399e3a00dac91e8d1 — 3047 bytes
- 06_TRACE_ME_BOUNDARY_STUB_v0_1.md —
406f38aec82e67df5ea93e66ebff0d4d1301f76cb15ad72cc119ee742aeecbc1 — 2303 bytes
- 07_TRACE_REVIEW_PROMPTS_v0_1.md —
33e7987bd2a0da6c4b2949f718da7facb0d208d9f89e418033ea3dc5fcf839be — 3072 bytes

Content bytes excluding manifest: 31399